# 8.18
# 汇报

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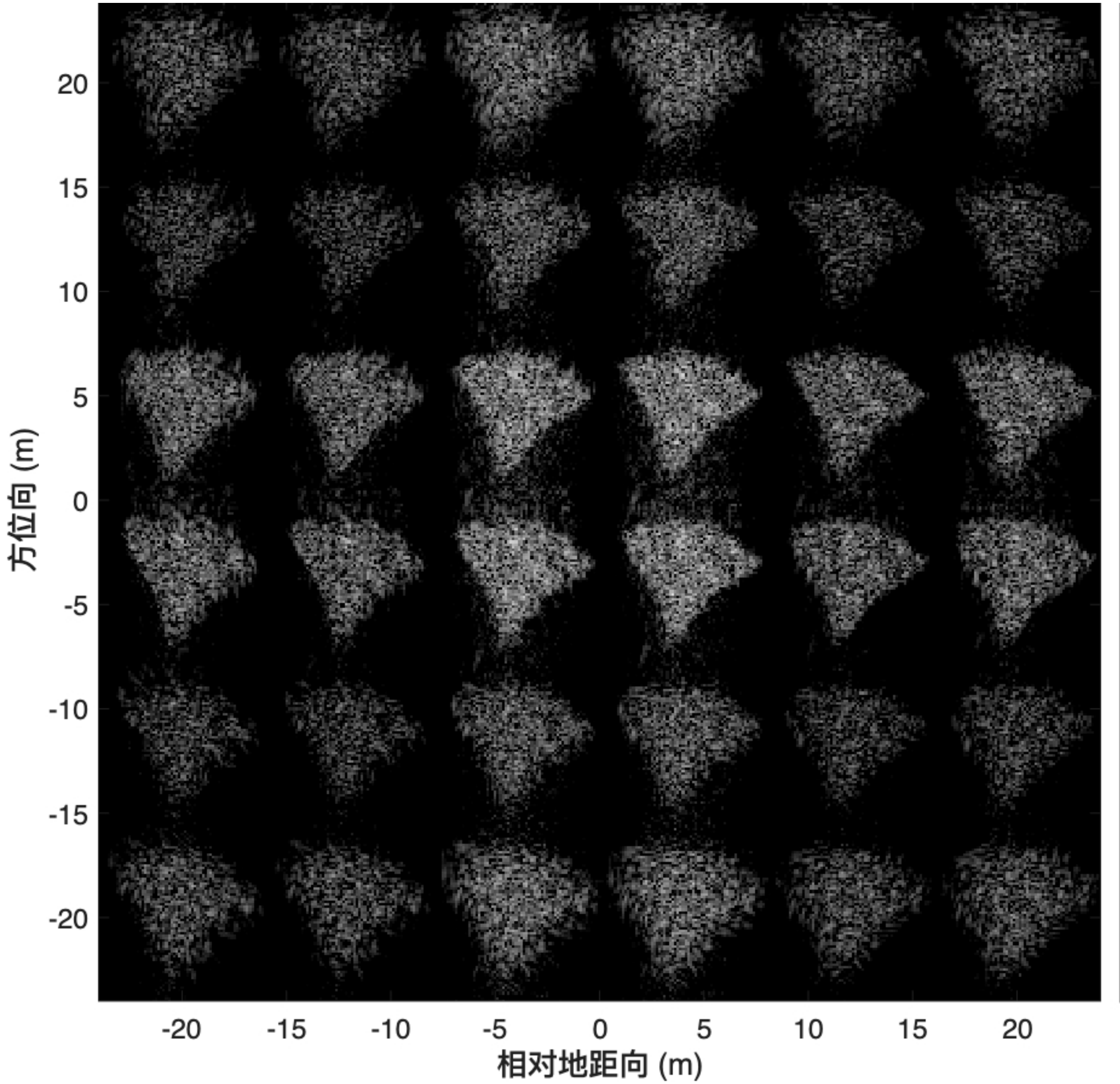
# 目录

CONTENTS

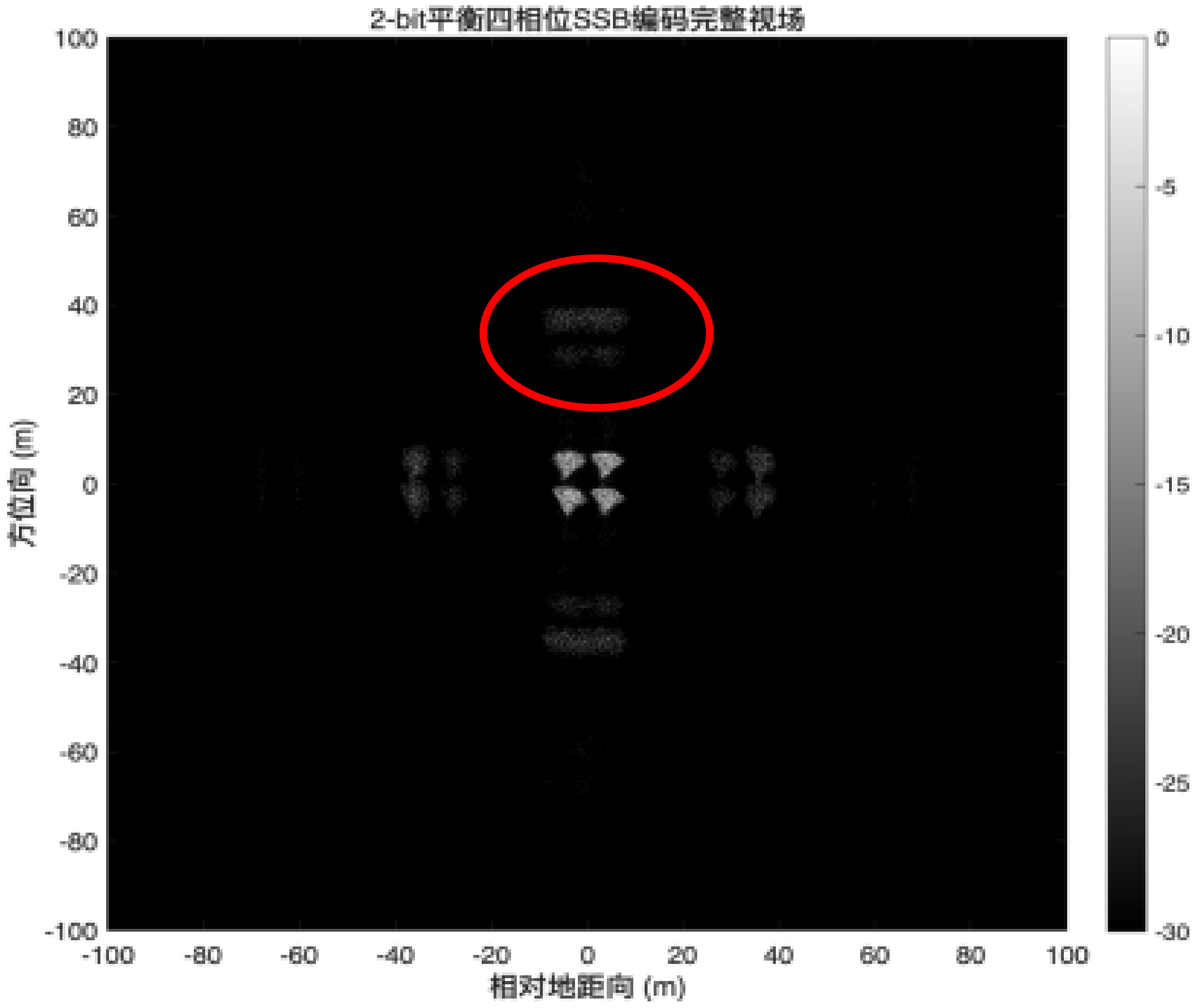
# 1 Meta Pencil自由度

**1bit超表面，无法抑制高次谐波对称谐波，效果差！！**

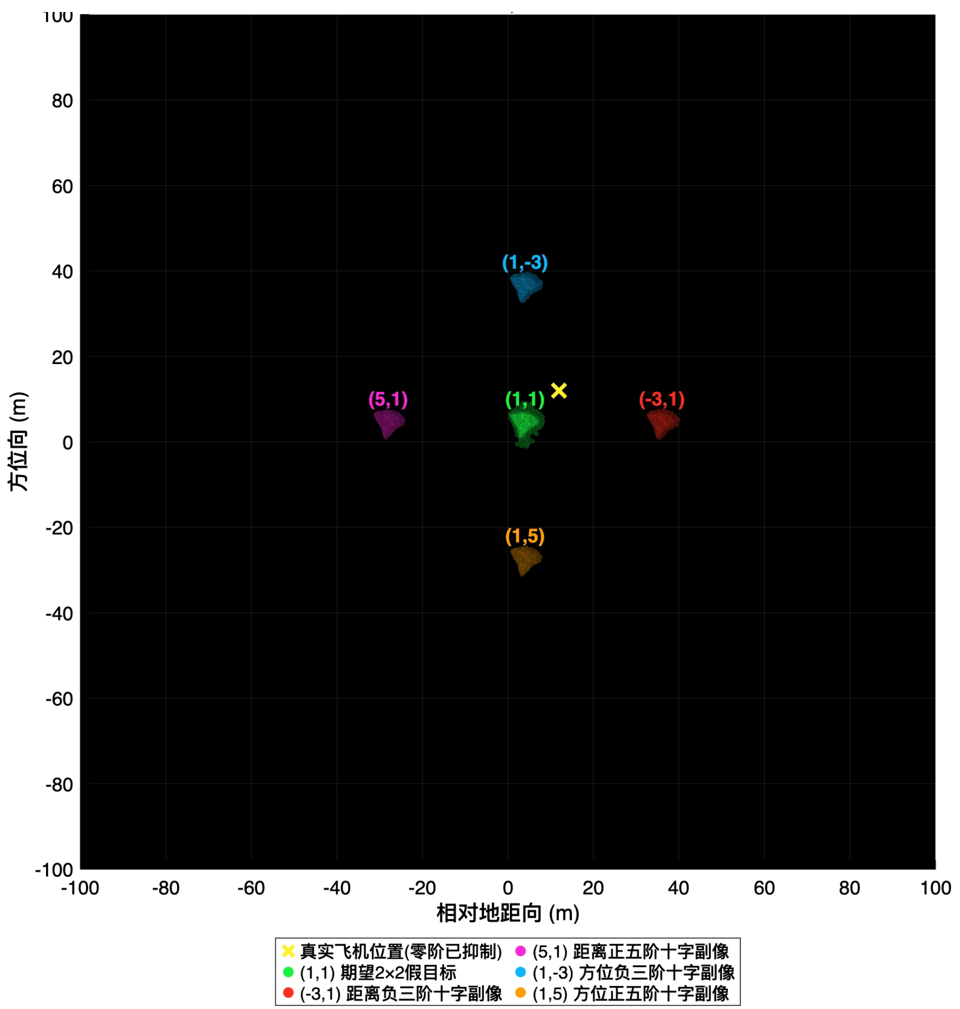
# 1 Meta Pencil自由度

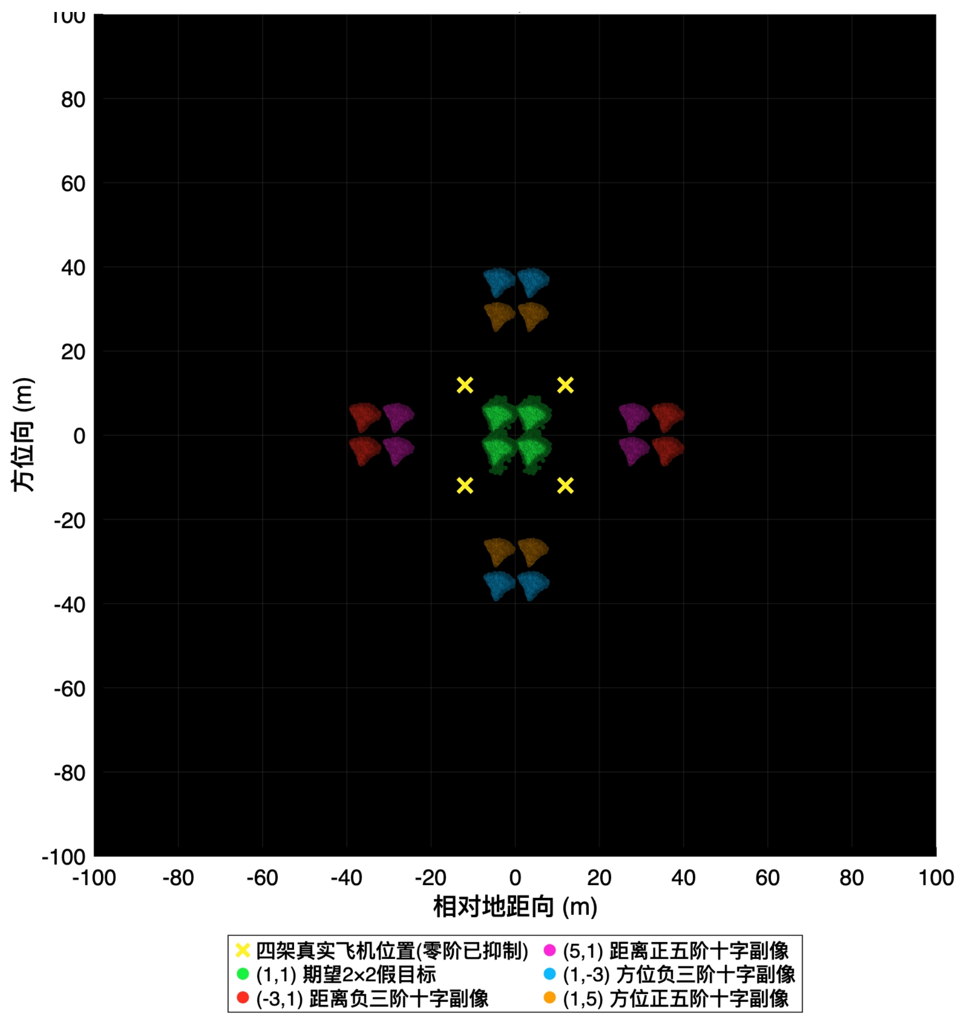
## 使用2bit超表面可以较好抑制高次谐波，但是还有外侧十字谐波



# 1 Matlab-飞机

- 成像结果中，会出现外侧十字的高阶谐波副像
  在2bit编码时，频谱会保留$n = \ldots, -7, -3, +1, +5, +9, \ldots$



# 1 Meta Pencil自由度

- 相关工作1：用的3-bit来改善谐波能量，模拟连续相位

Open Access Article

**Variable Frequency Phase Modulation on Time-Modulated Metasurface for SAR Feature Reconstruction**

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Key Laboratory of Complex Electromagnetic Environment Effects on Electronics and Information System, College of Electronic Science and Technology, National University of Defense Technology, Changsha 410073, China

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**Highlights**

**What are the main findings?**

- A SAR feature reconstruction method based on variable frequency phase modulation is proposed, enabling a single metasurface to modulate multiple scattering centers in both range and azimuth dimensions.
- An analytical inverse model is established to link modulation parameters with the spatial coordinates and amplitudes of generated scattering centers, allowing for the simultaneous reconstruction of target features through independent design of modulation duration and frequency.

**What are the implications of the main findings?**

- This study marks the first systematic application of variable frequency-modulation techniques to SAR feature reconstruction, breaking through the inherent limitations of traditional one-to-one mapping. It provides a novel technical solution for achieving efficient, flexible, and high-fidelity simulation of complex target electromagnetic characteristics.
- An inverse analytical expression has been established between the target electromagnetic scattering characteristics and the time-domain modulation parameters of the metasurface. Through independent design of the modulation duration and modulation frequency, simultaneous reconstruction of the position and amplitude information of the generated scattering points is achieved.

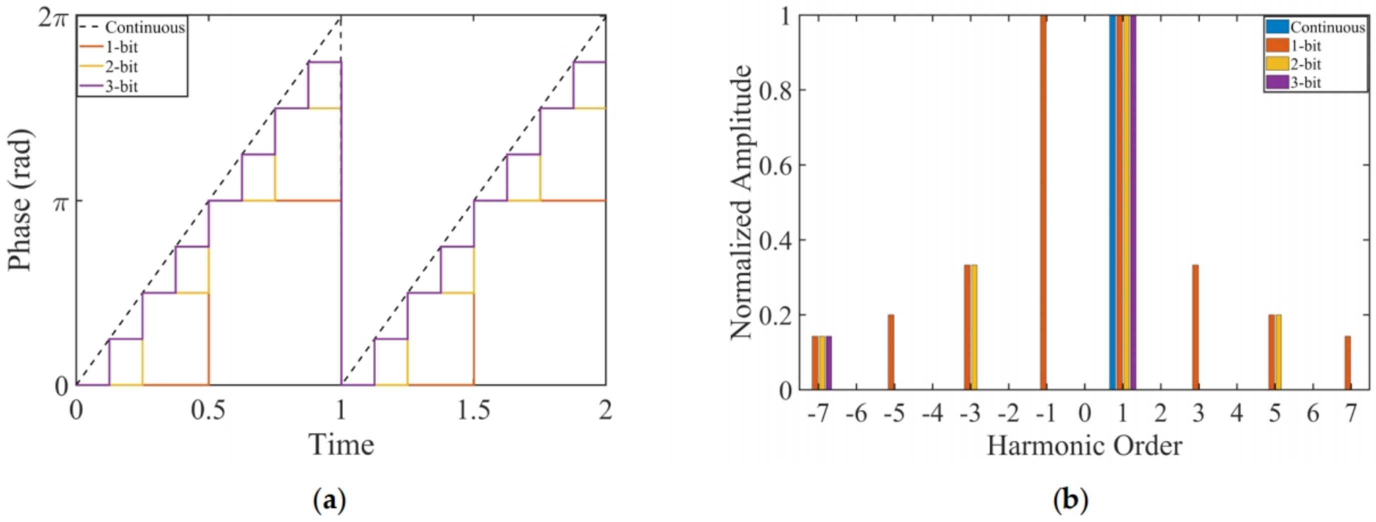



**Figure 3.** Comparison of 1-bit, 2-bit, 3-bit, and continuous linear-phase modulation. (**a**) Modulation waveforms; (**b**) Harmonic order and normalized amplitude.

bit数越高，能量在1阶越集中

# 1 Meta Pencil自由度

- 相关工作1：用的3-bit来改善谐波能量，模拟连续相位

这篇文章里面有很多SAR成像图

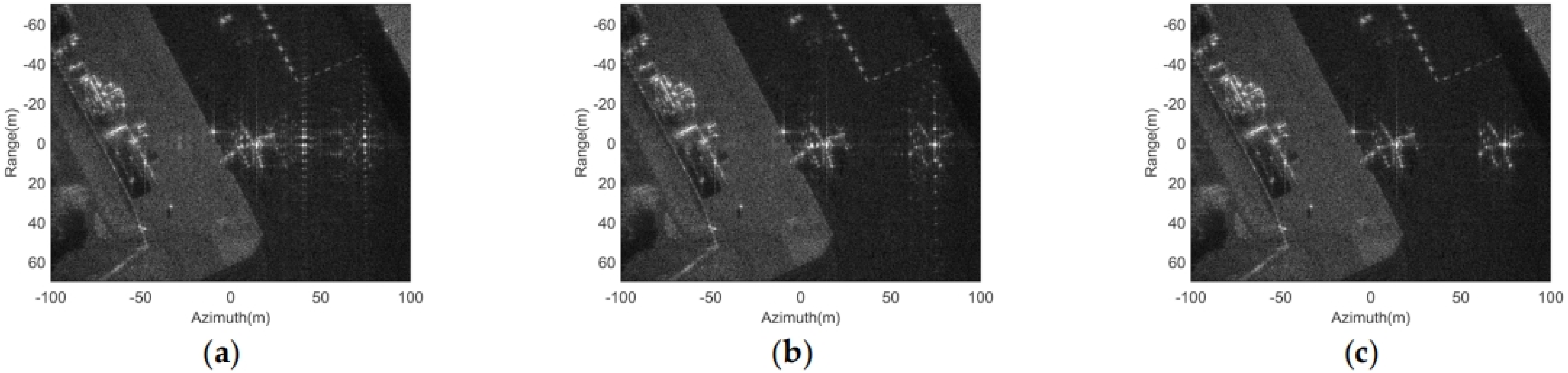



**Figure 17.** Comparison of SAR images of the reconstructed aircraft target and the original aircraft target under different phase quantization bits. **(a)** 1-bit; **(b)** 2-bit; **(c)** continuous phase modulation.

# 1 Meta Pencil自由度

- 相关工作2：空时联合编码
- 为不同子区设计不同时间序列、起始相位和空间相位
- 各个分区之间是异步的

1. 时间延迟给k阶谐波添加k倍相位

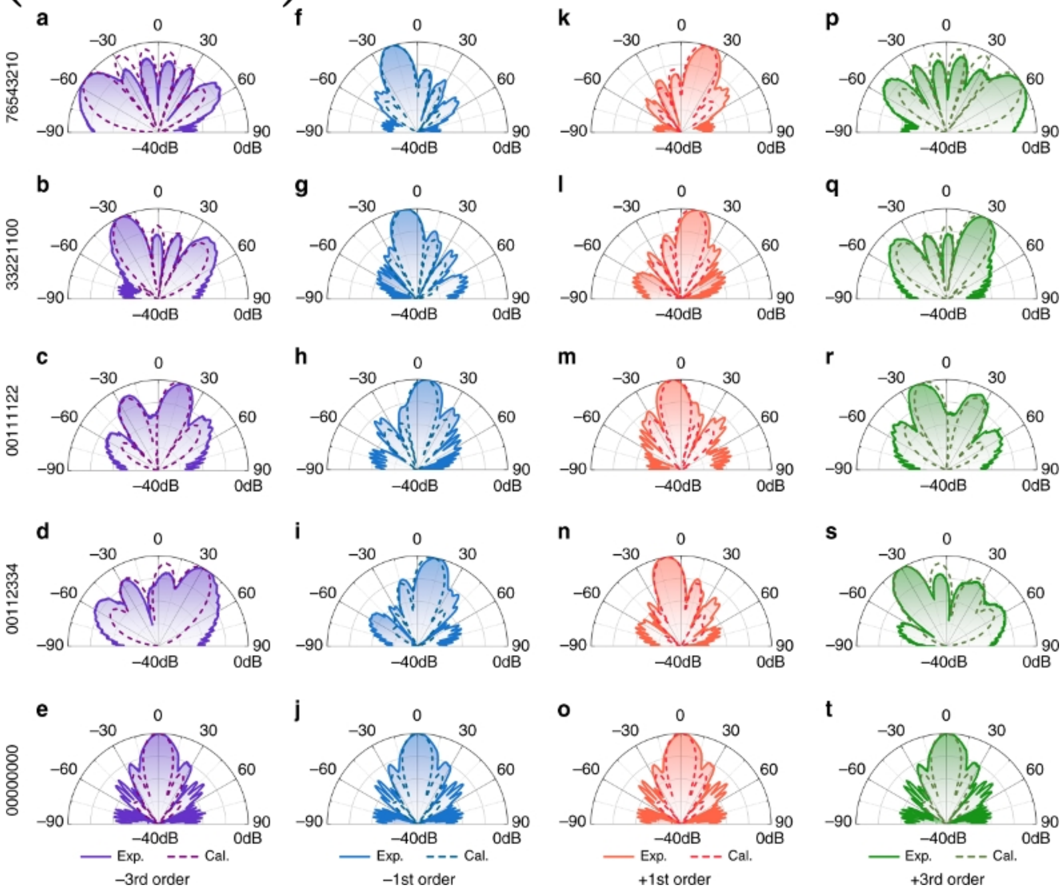
$$\Gamma(t - t_0) \quad \Longrightarrow \quad e^{-jk\omega_0 t_0}$$

$$E_r(\omega) = 2\pi A\cos\frac{\phi_2-\phi_1}{2}e^{j\frac{\phi_2+\phi_1}{2}}E_i(\omega) + \sum_{m=-\infty}^{+\infty}\frac{4A}{2m-1}\sin\frac{\phi_2-\phi_1}{2}e^{j\left[\frac{\phi_2+\phi_1}{2}-(2m-1)\omega_0 t_0\right]} E_i[\omega-(2m-1)\omega_0] \quad (7)$$

Specifically, the amplitude and phase of the $k^{\mathrm{th}}$ order harmonic are rewritten as

$$E_r(k\omega_0+\omega_c) = |A|\angle\Phi = \begin{cases} \left|A\cos\frac{\phi_2-\phi_1}{2}\right| e^{j\left\{\frac{\phi_2+\phi_1}{2}+\pi\left[\varepsilon\left(\frac{\phi_2-\phi_1}{2}-\pi\right)+\varepsilon\left(\frac{\phi_1-\phi_2}{2}-\pi\right)\right]\right\}} & k=0 \\ \left|\frac{2A}{k\pi}\sin\frac{\phi_2-\phi_1}{2}\right| e^{j\left\{\frac{\phi_2+\phi_1}{2}-\pi[\varepsilon(\phi_2-\phi_1)+\varepsilon(k)]-k\omega_0 t_0\right\}} & k=\pm1,\pm3,\pm5\cdots \\ 0 & k=\pm2,\pm4,\pm6\cdots \end{cases} \quad (8)$$

The extra phase shift provides new degrees of freedom to control the propagation features of high-order harmonics by breaking the constraint between their amplitude and phase, making it possible to realize independent regulation of both parameters, which is hard to achieve using conventional methods. In fact, Eq. (8) offers a promising route to tailor the response of reflected



# 1 Meta Pencil自由度

## 2. 交织子序列把2-bit时间码变成可设计的谐波向量

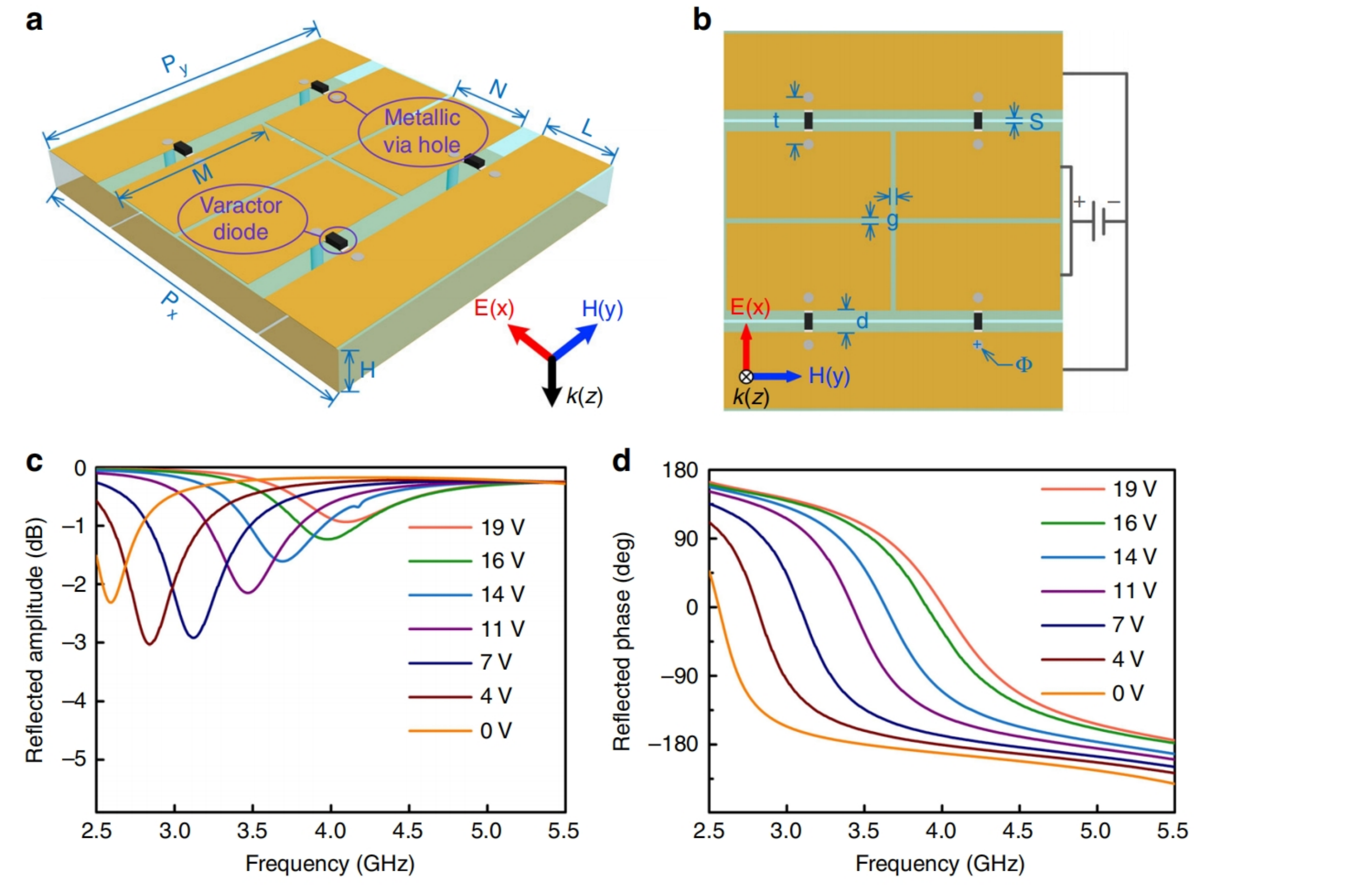
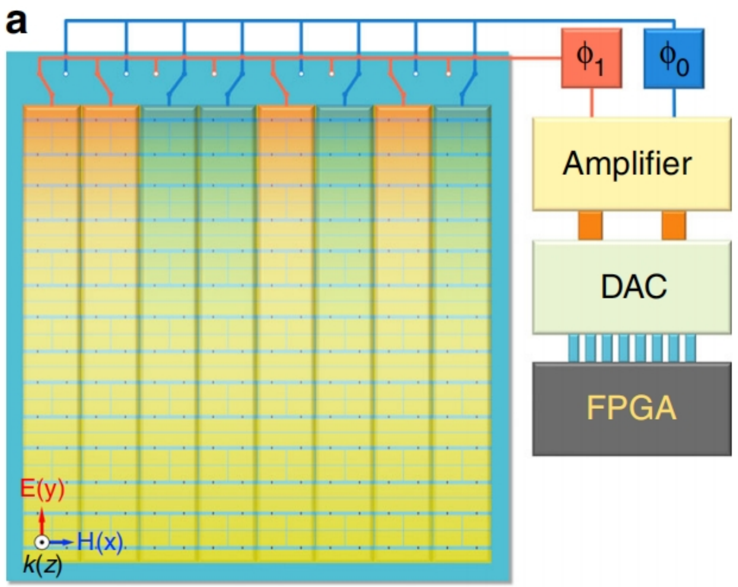


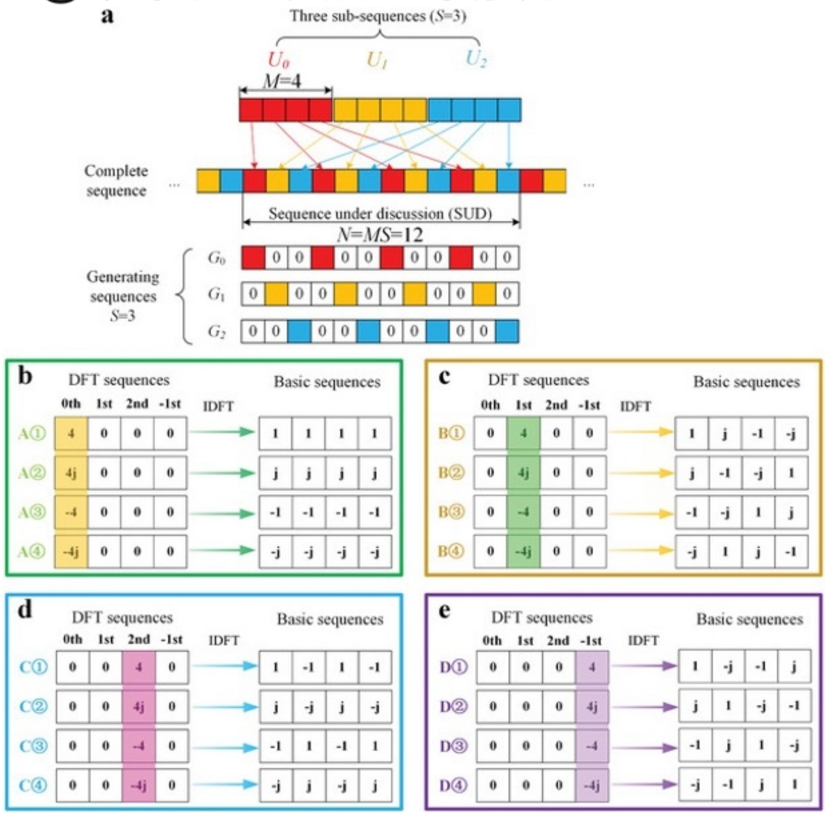

**Fig. 2 Illustration of the time-domain digital coding meta-atom. a**, **b** Perspective and front views of the time-domain digital coding meta-atom. **c**, **d** The simulated reflection amplitude (**c**) and phase (**d**) spectra under different bias voltages. The dimensions of the meta-atom are $P_x = 37$ mm, $P_y = 33$ mm, $H = 4$ mm, $M = 16$ mm, $N = 8.45$ mm, $L = 7.45$ mm, $g = 0.5$ mm, $d = 2.1$ mm, $t = 4.6$ mm, $\Phi = 1$ mm, $s = 0.3$ mm



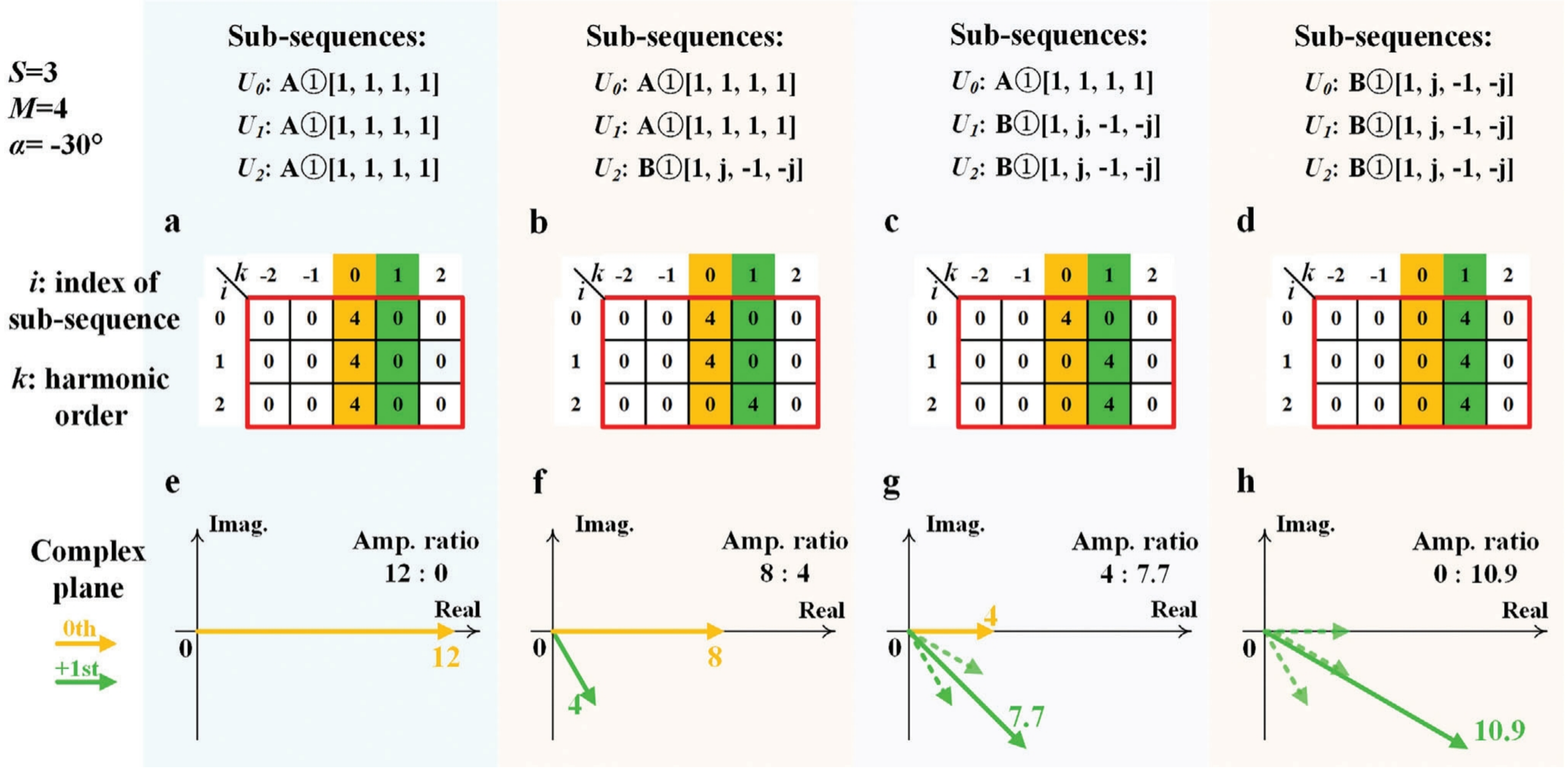
# 1 Meta Pencil自由度

## 2. 交织子序列把2-bit时间码变成可设计的谐波向量

### ① 先构造正交的基本序列




### ② 再把各子序列贡献做复数矢量叠加



# 1 Meta Pencil自由度

## 优化思路

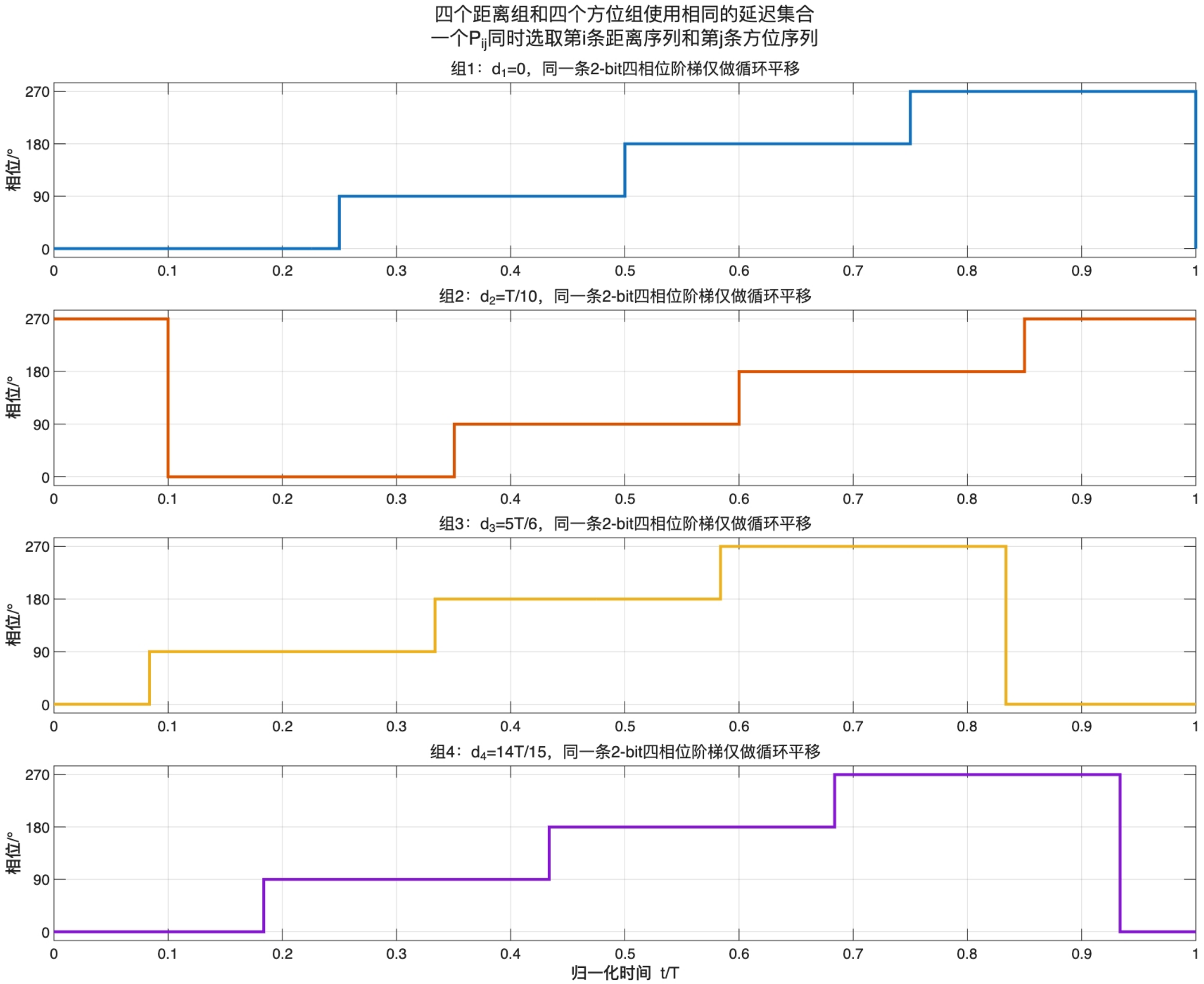
一个局部4×4超单元：$P_{ij}=R_i\times A_j$

每格仍是单极化2-bit四相位单元，不是固定相位块

方位向时延组 $A_j$

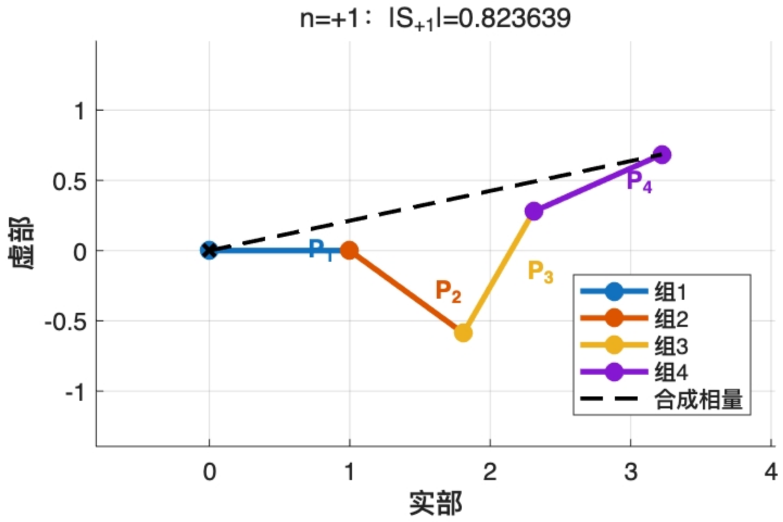
距离向时延组 $R_i$

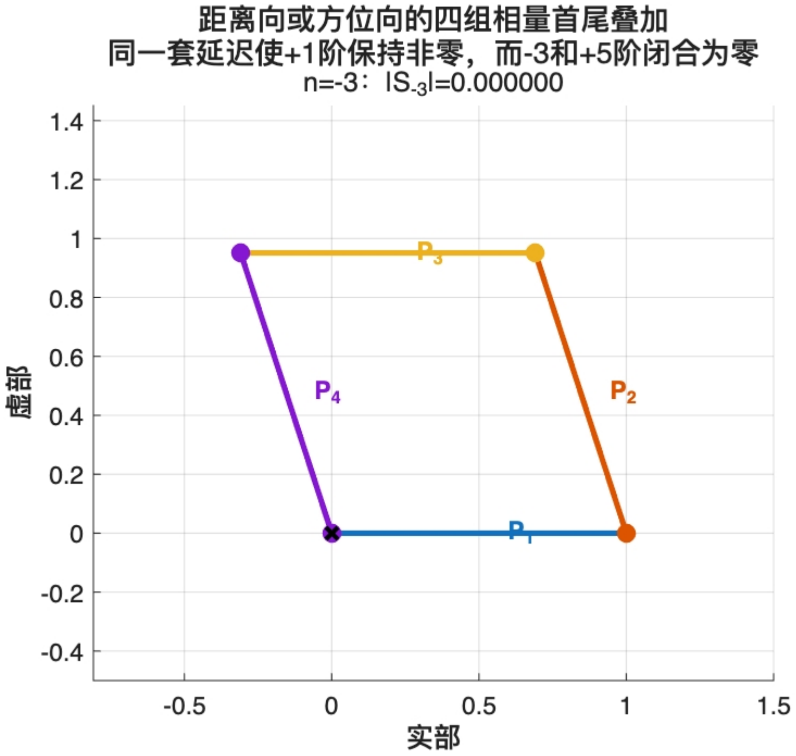
| | $A_1$ | $A_2$ | $A_3$ | $A_4$ |
|---|---|---|---|---|
| $R_1$ | **$P_{11}$** $d_r=0$ $d_a=0$ | **$P_{12}$** $d_r=0$ $d_a=T/10$ | **$P_{13}$** $d_r=0$ $d_a=5T/6$ | **$P_{14}$** $d_r=0$ $d_a=14T/15$ |
| $R_2$ | **$P_{21}$** $d_r=T/10$ $d_a=0$ | **$P_{22}$** $d_r=T/10$ $d_a=T/10$ | **$P_{23}$** $d_r=T/10$ $d_a=5T/6$ | **$P_{24}$** $d_r=T/10$ $d_a=14T/15$ |
| $R_3$ | **$P_{31}$** $d_r=5T/6$ $d_a=0$ | **$P_{32}$** $d_r=5T/6$ $d_a=T/10$ | **$P_{33}$** $d_r=5T/6$ $d_a=5T/6$ | **$P_{34}$** $d_r=5T/6$ $d_a=14T/15$ |
| $R_4$ | **$P_{41}$** $d_r=14T/15$ $d_a=0$ | **$P_{42}$** $d_r=14T/15$ $d_a=T/10$ | **$P_{43}$** $d_r=14T/15$ $d_a=5T/6$ | **$P_{44}$** $d_r=14T/15$ $d_a=14T/15$ |

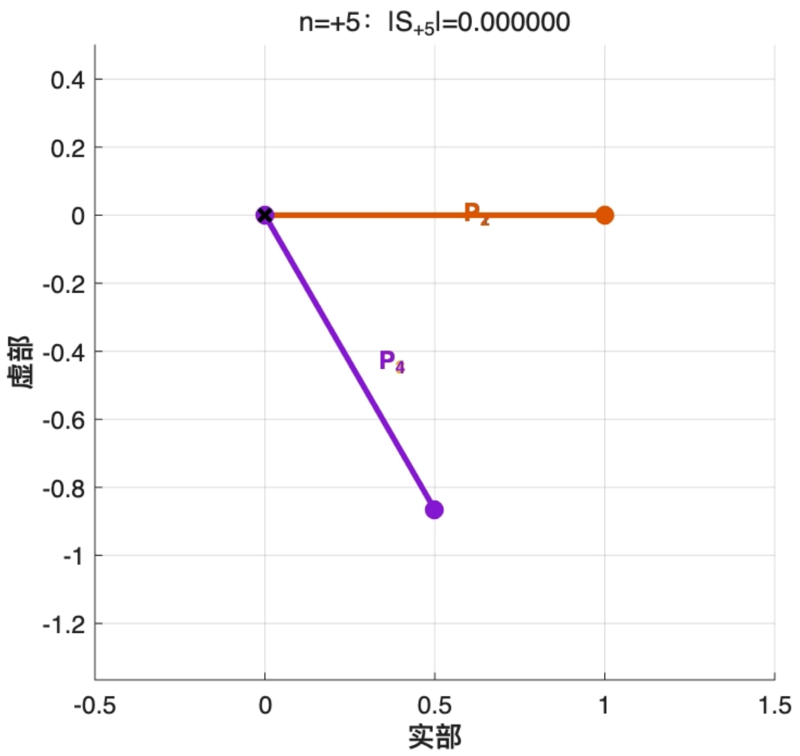


# 1 Meta Pencil自由度

## 优化思路



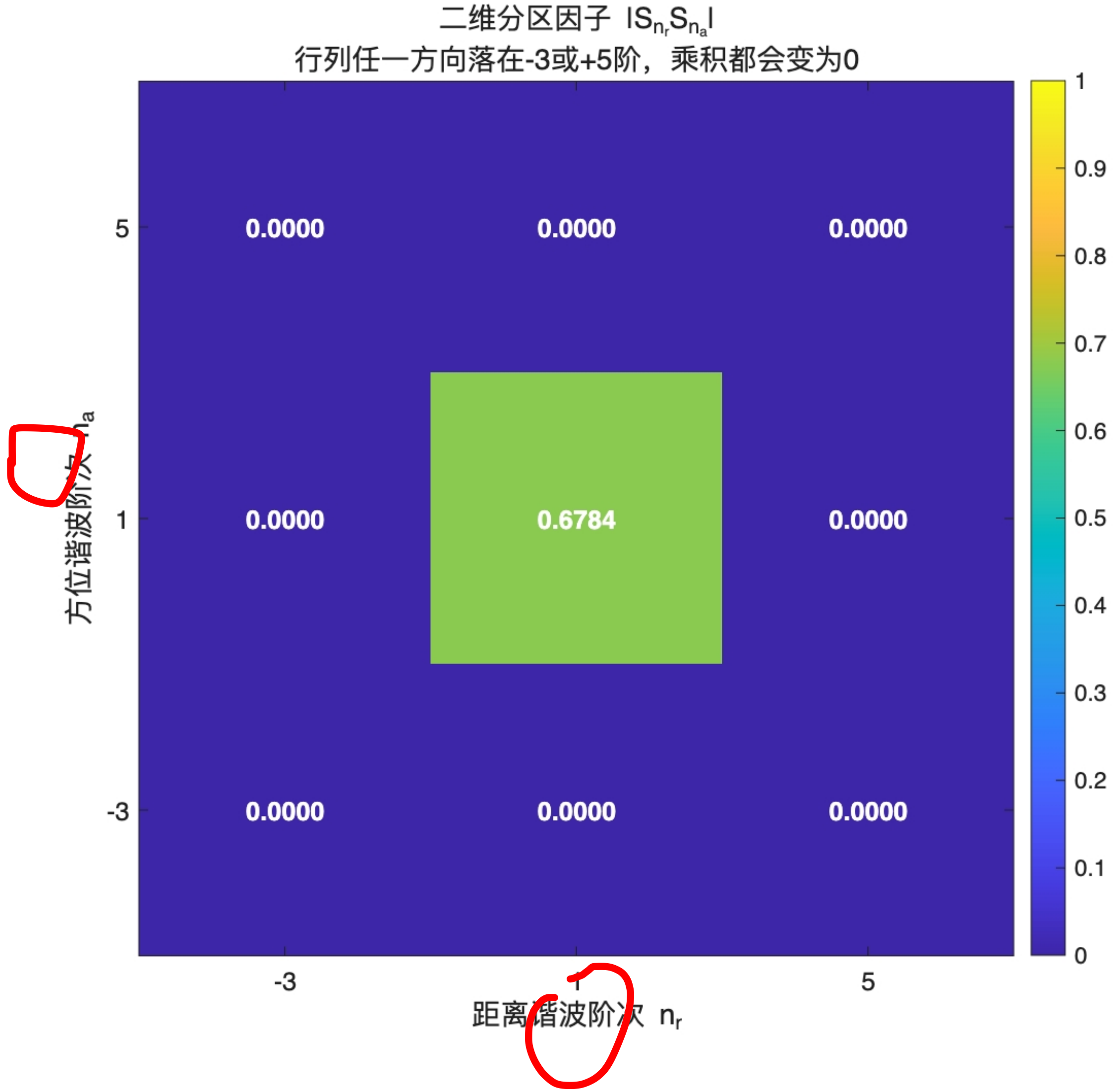

+1阶谐波，向量和不为0

-3阶谐波，向量和为0

+5阶谐波，向量和为0

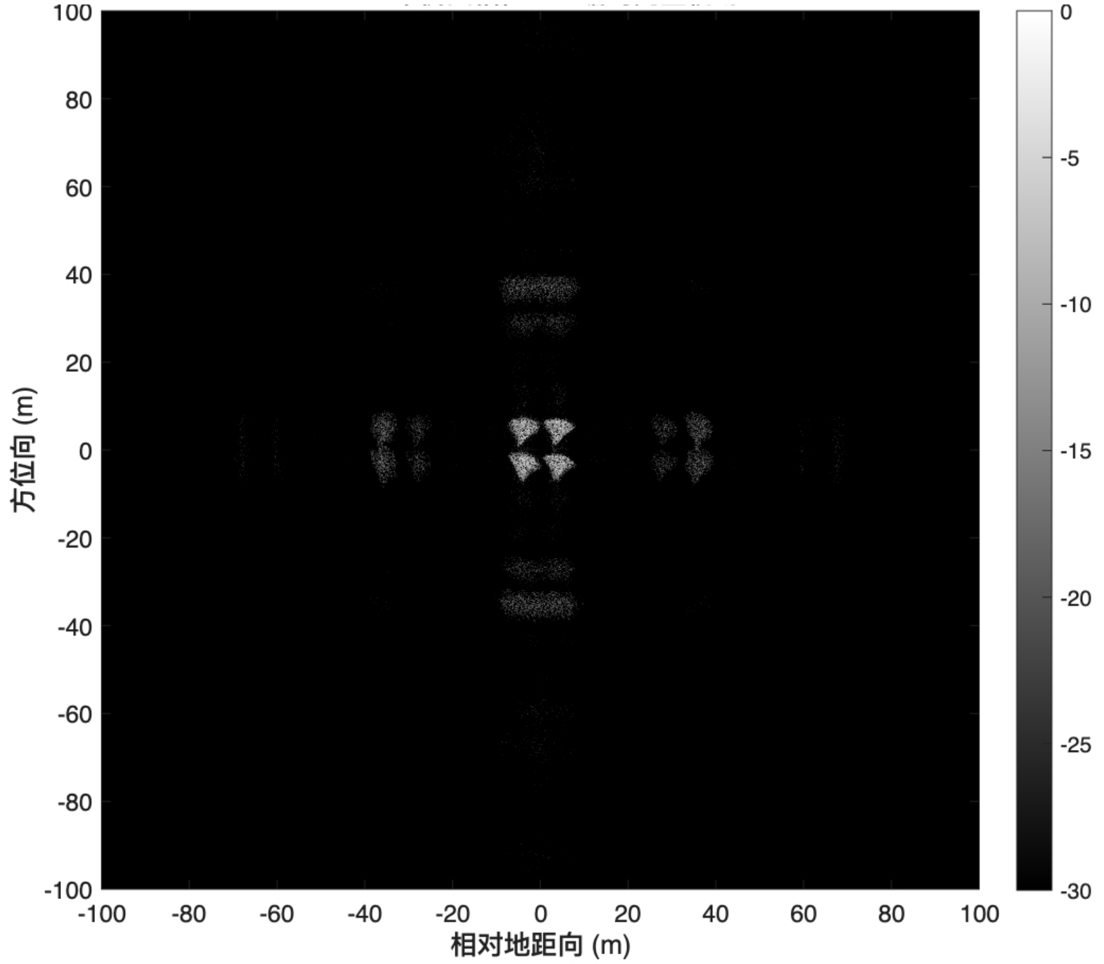
# 1 Meta Pencil自由度

## 只有方位向和距离向的1阶谐波有能量，相邻阶次均被抑制

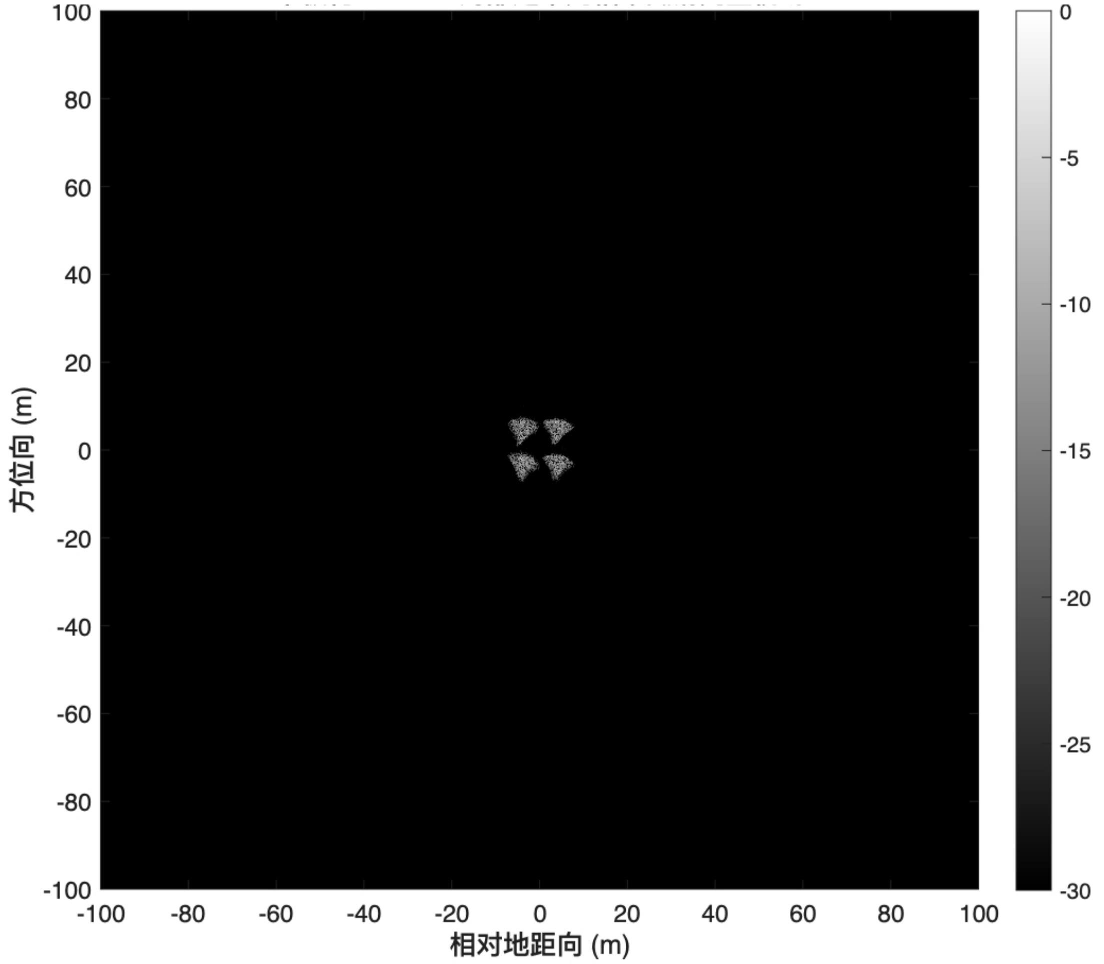


# 1 Meta Pencil自由度

## 优化思路




优化前




优化后

# 1 Meta Pencil自由度

使用这种思路，可以做出很多高质量的电磁拟像
整体结构如下：

## Hk内部：4×4延迟超表面单元

行：距离延迟　　列：方位延迟

| | A1 $d_a$=0T | A2 $d_a$=1/10T | A3 $d_a$=5/6T | A4 $d_a$=14/15T |
|---|---|---|---|---|
| R1 $d_r$=0T | P11 (0, 0)T | P12 (0, 1/10)T | P13 (0, 5/6)T | P14 (0, 14/15)T |
| R2 $d_r$=1/10T | P21 (1/10, 0)T | P22 (1/10, 1/10)T | P23 (1/10, 5/6)T | P24 (1/10, 14/15)T |
| R3 $d_r$=5/6T | P31 (5/6, 0)T | P32 (5/6, 1/10)T | P33 (5/6, 5/6)T | P34 (5/6, 14/15)T |
| R4 $d_r$=14/15T | P41 (14/15, 0)T | P42 (14/15, 1/10)T | P43 (14/15, 5/6)T | P44 (14/15, 14/15)T |

空间编码的第一部分

### 1　目标偏移 → 调制频率

$\Delta r_1 = +8$ m, $\Delta R = \Delta r_1 \sin 60° = 6.928$ m

$f_{r,1} = -2K_r\Delta R/c = -32$ MHz

$\Delta a_1 = +8$ m, $f_{a,1} = -K_a\Delta a_1/v = +42.67$ Hz

### 2　选择超表面的距离/方位时延

$T_r = 1/|f_{r,1}| = 31.25$ ns

$d_r = 1/10 \Rightarrow t_{delay,r} = 3.125$ ns

$T_a = 1/|f_{a,1}| = 23.438$ ms

$d_a = 5/6 \Rightarrow t_{delay,a} = 19.531$ ms

### 3　选取一个快/慢时间采样点，分别计算连续相位

$t_{fast} = 0.30T_r = 9.375$ ns $t_{slow} = 0.20T_a = 4.688$ ms

$\varphi_r = 360°[-0.30 - 0.10] = -144°$

$\varphi_a = 360°[+0.20 - 5/6] = -228°$

### 4　量化为四种相位状态并相乘

$Q_2(216°) = \mathbf{180°\ (10)}$ $Q_2(132°) = \mathbf{90°\ (01)}$

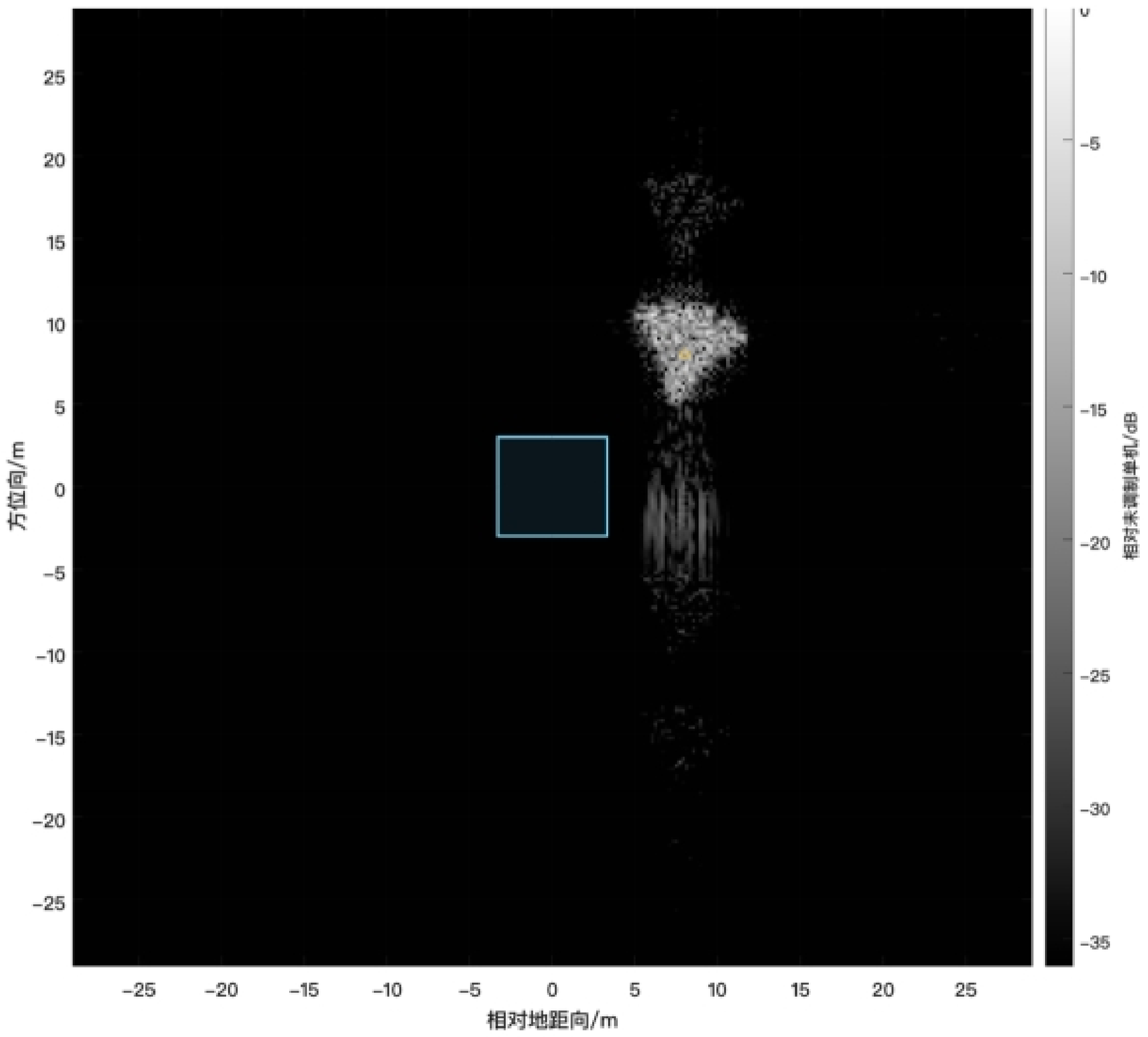
$\varphi_{out} = (180° + 90°) \bmod 360° = \mathbf{270°} \Rightarrow$ 2-bit = **11**

# 1 Meta Pencil自由度

## 1 单个飞机

### 1.1 单假目标

| | A1 | A2 | A3 | A4 |
|---|---|---|---|---|
| R1 | H1 (+8, +8) m | H1 (+8, +8) m | H1 (+8, +8) m | H1 (+8, +8) m |
| R2 | H1 (+8, +8) m | H1 (+8, +8) m | H1 (+8, +8) m | H1 (+8, +8) m |
| R3 | H1 (+8, +8) m | H1 (+8, +8) m | H1 (+8, +8) m | H1 (+8, +8) m |
| R4 | H1 (+8, +8) m | H1 (+8, +8) m | H1 (+8, +8) m | H1 (+8, +8) m |



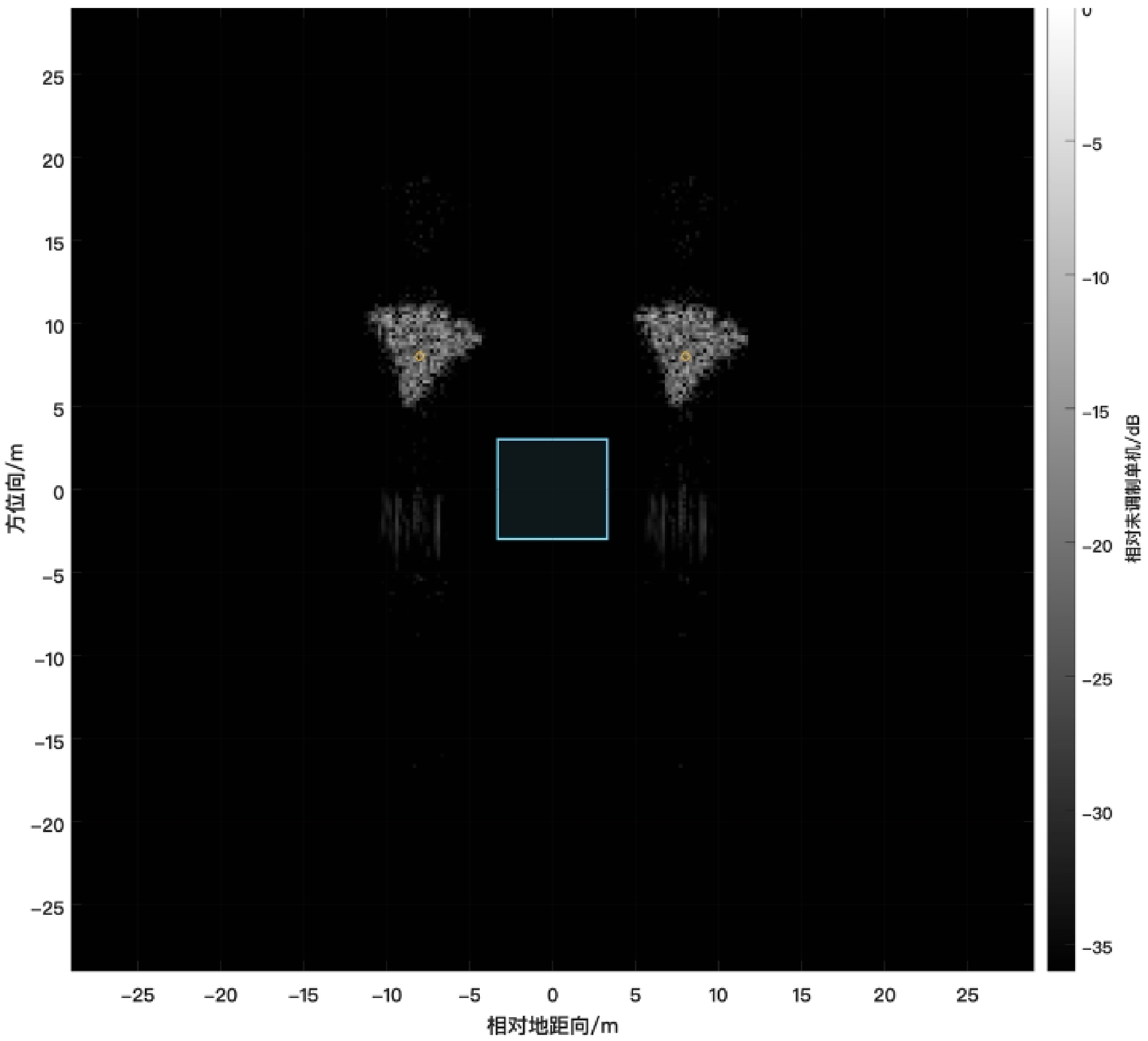
# 1 Meta Pencil自由度

## 1.2 双假目标

| | A1 | A2 | A3 | A4 |
|---|---|---|---|---|
| R1 | H1 (-8, +8) m | H2 (+8, +8) m | H1 (-8, +8) m | H2 (+8, +8) m |
| R2 | H2 (+8, +8) m | H1 (-8, +8) m | H2 (+8, +8) m | H1 (-8, +8) m |
| R3 | H1 (-8, +8) m | H2 (+8, +8) m | H1 (-8, +8) m | H2 (+8, +8) m |
| R4 | H2 (+8, +8) m | H1 (-8, +8) m | H2 (+8, +8) m | H1 (-8, +8) m |

H1: (-8,+8) m ×8　　H2: (+8,+8) m ×8

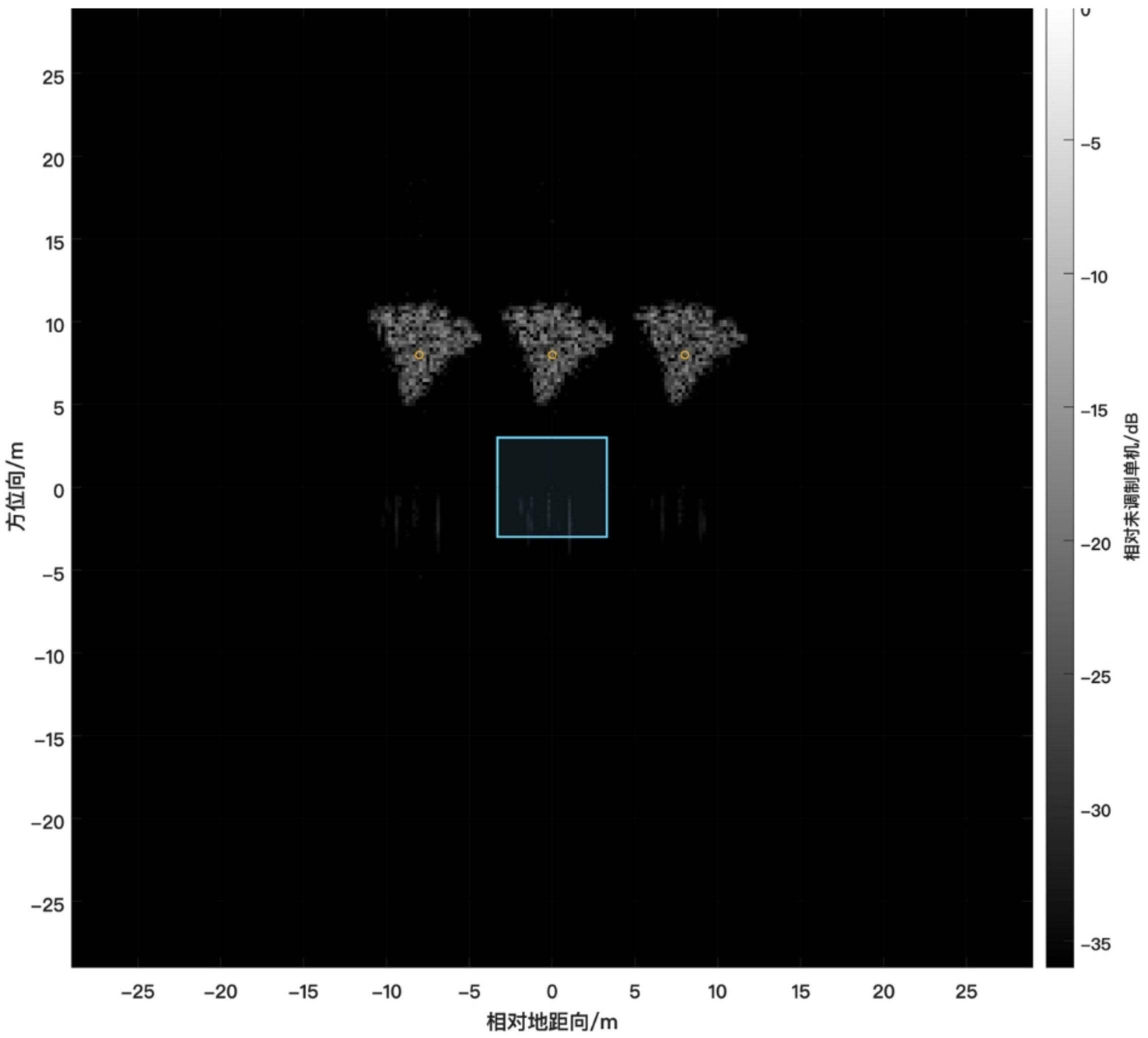
H1:（-8，+8）
H2:（+8，+8）



# 1 Meta Pencil自由度

## 1.3 三假目标

| | A1 | A2 | A3 | A4 | A5 |
|---|---|---|---|---|---|
| R1 | H1 (-8, +8) m | H2 (+0, +8) m | H3 (+8, +8) m | H1 (-8, +8) m | H2 (+0, +8) m |
| R2 | H1 (-8, +8) m | H3 (+8, +8) m | H2 (+0, +8) m | H1 (-8, +8) m | H3 (+8, +8) m |
| R3 | H2 (+0, +8) m | H3 (+8, +8) m | H1 (-8, +8) m | H2 (+0, +8) m | H3 (+8, +8) m |
| R4 | H2 (+0, +8) m | H1 (-8, +8) m | H3 (+8, +8) m | H2 (+0, +8) m | H1 (-8, +8) m |
| R5 | H3 (+8, +8) m | H1 (-8, +8) m | H2 (+0, +8) m | H3 (+8, +8) m | H1 (-8, +8) m |



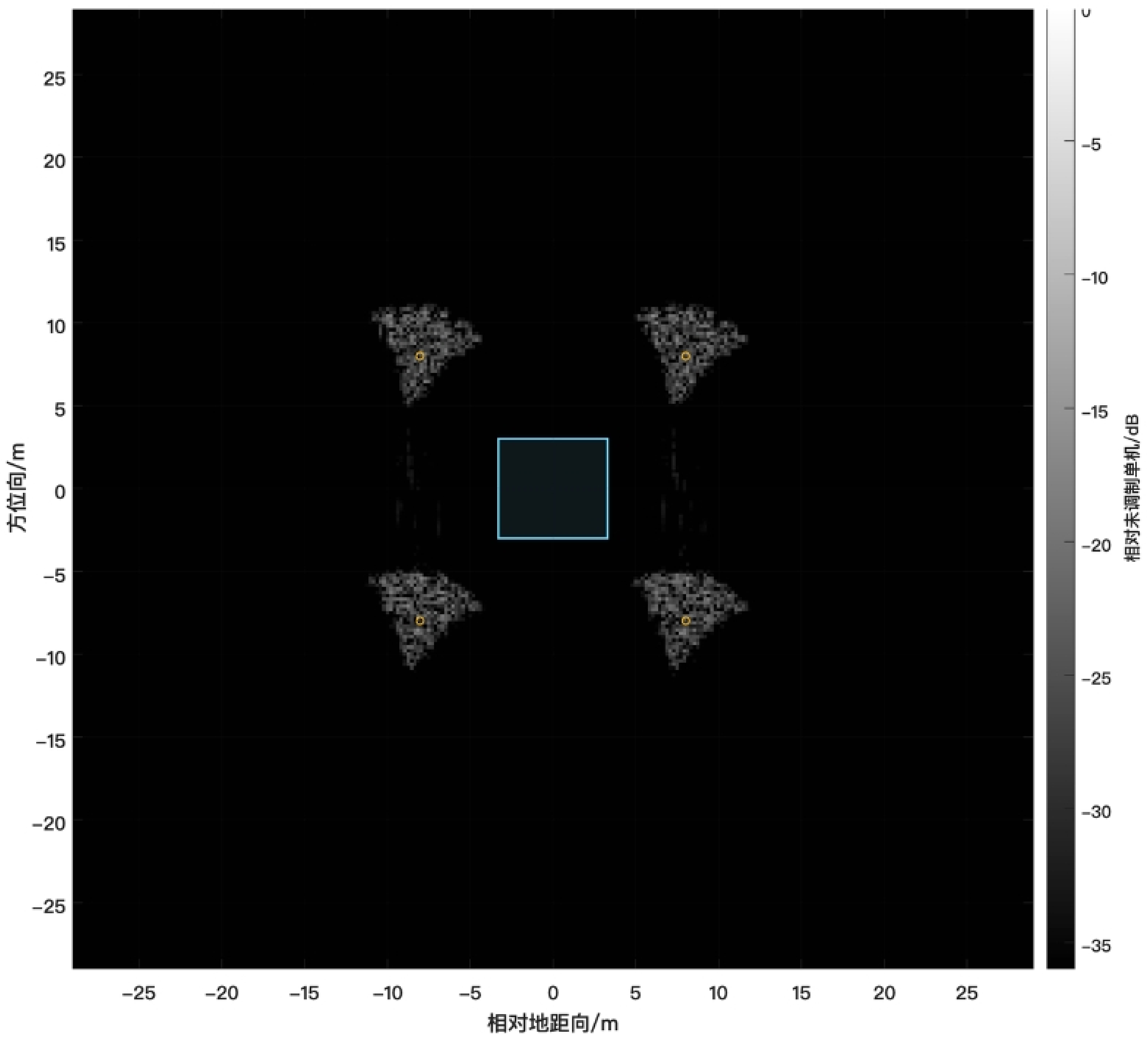

H1: (-8,+8) m ×9

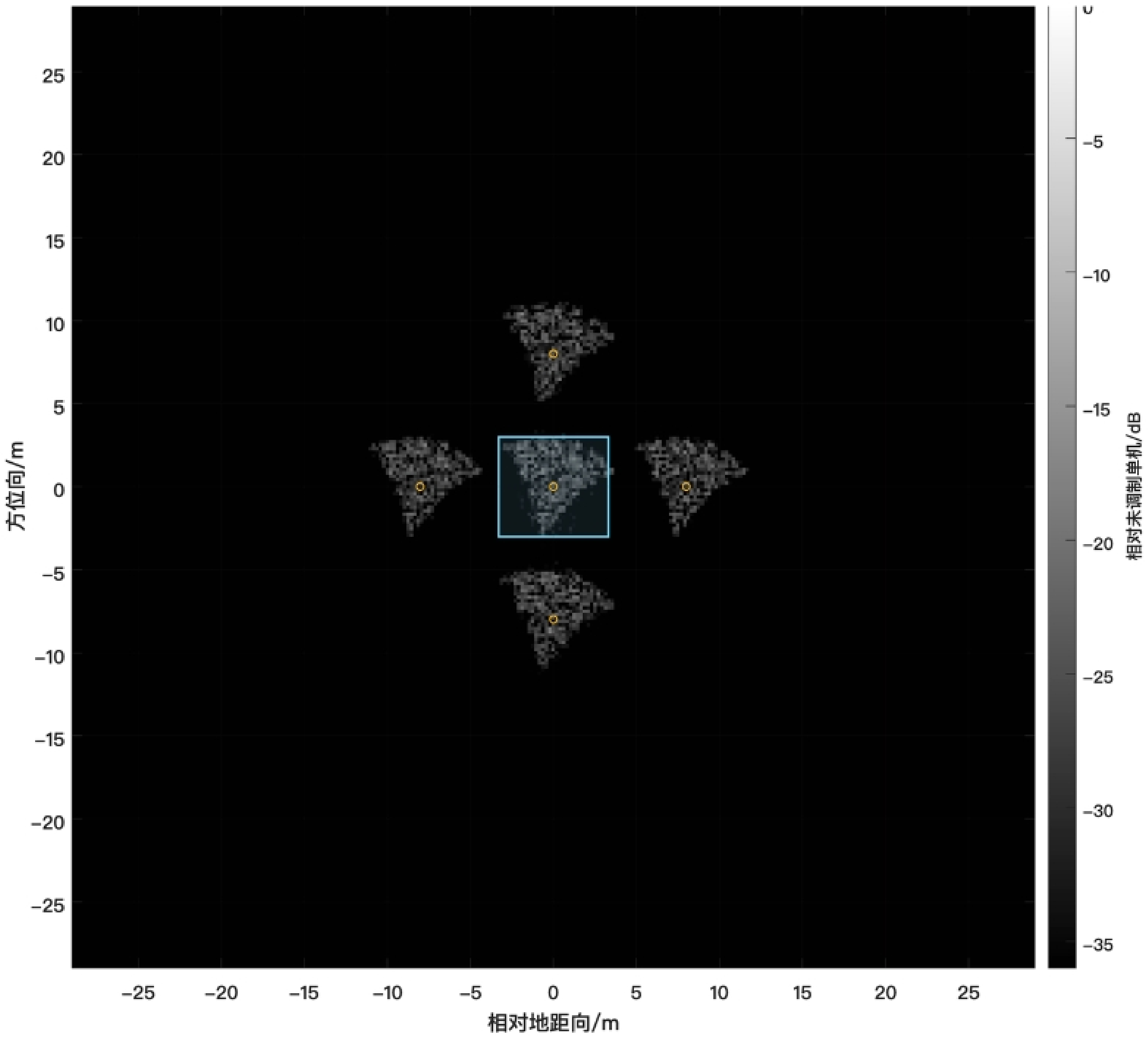
H2: (+0,+8) m ×8

H3: (+8,+8) m ×8

# 1 Meta Pencil自由度

## 1.4 四、五点假目标



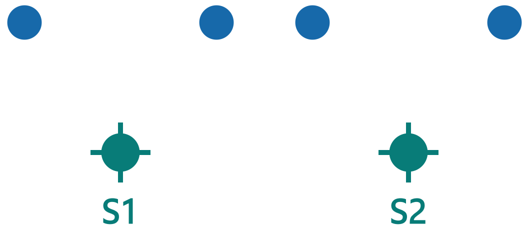



# 1 Meta Pencil自由度

## 2 两架飞机

### 共享码本

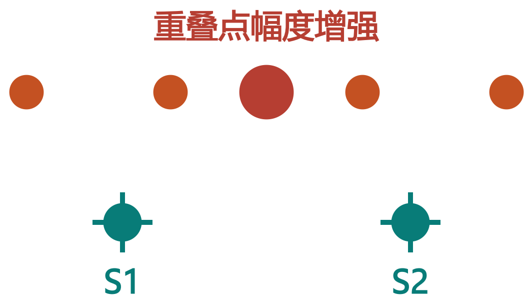
K1 = K2




平移到两架飞机处
规则重复

### 相干重叠

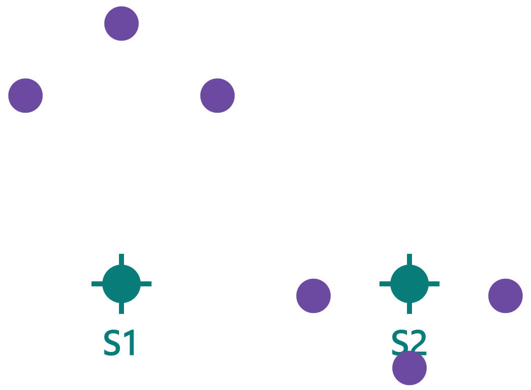
S1+K1 与 S2+K2 重合




重叠点接收两路复回波
可增强指定假目标

### 独立码本

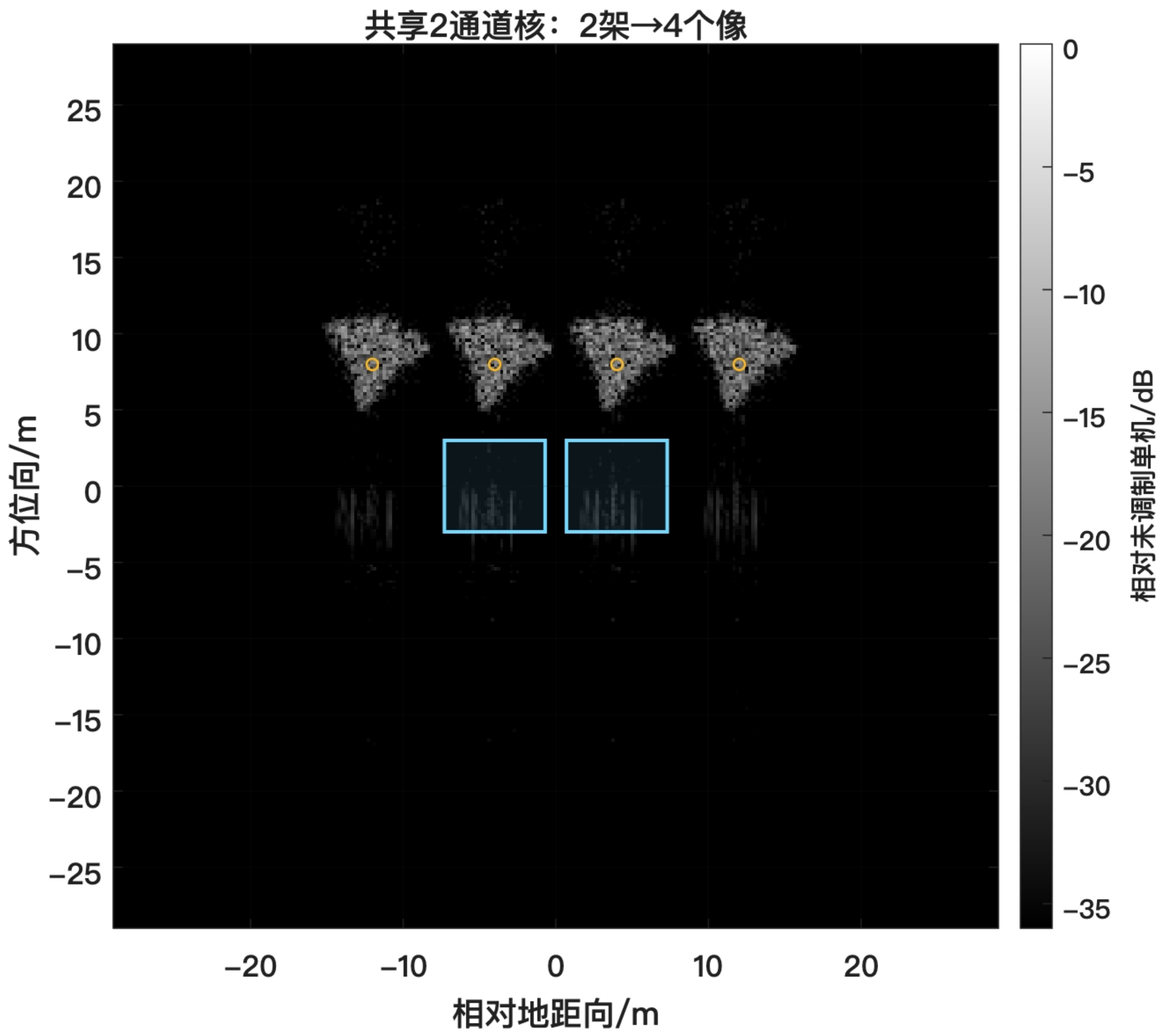
K1 ≠ K2



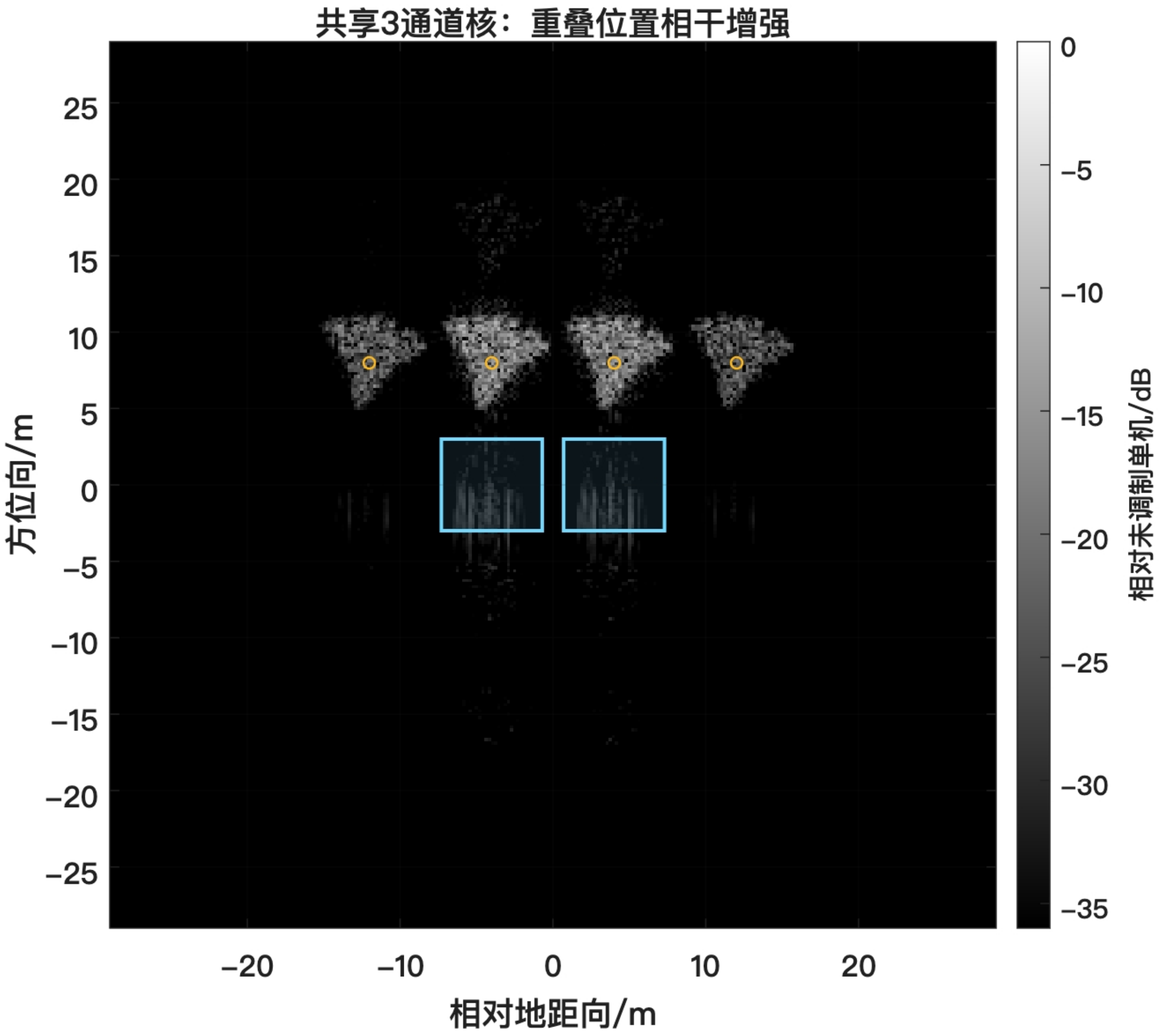

每架飞机生成不同子编队
可做非对称

# 1 Meta Pencil自由度

## 2.1 共享

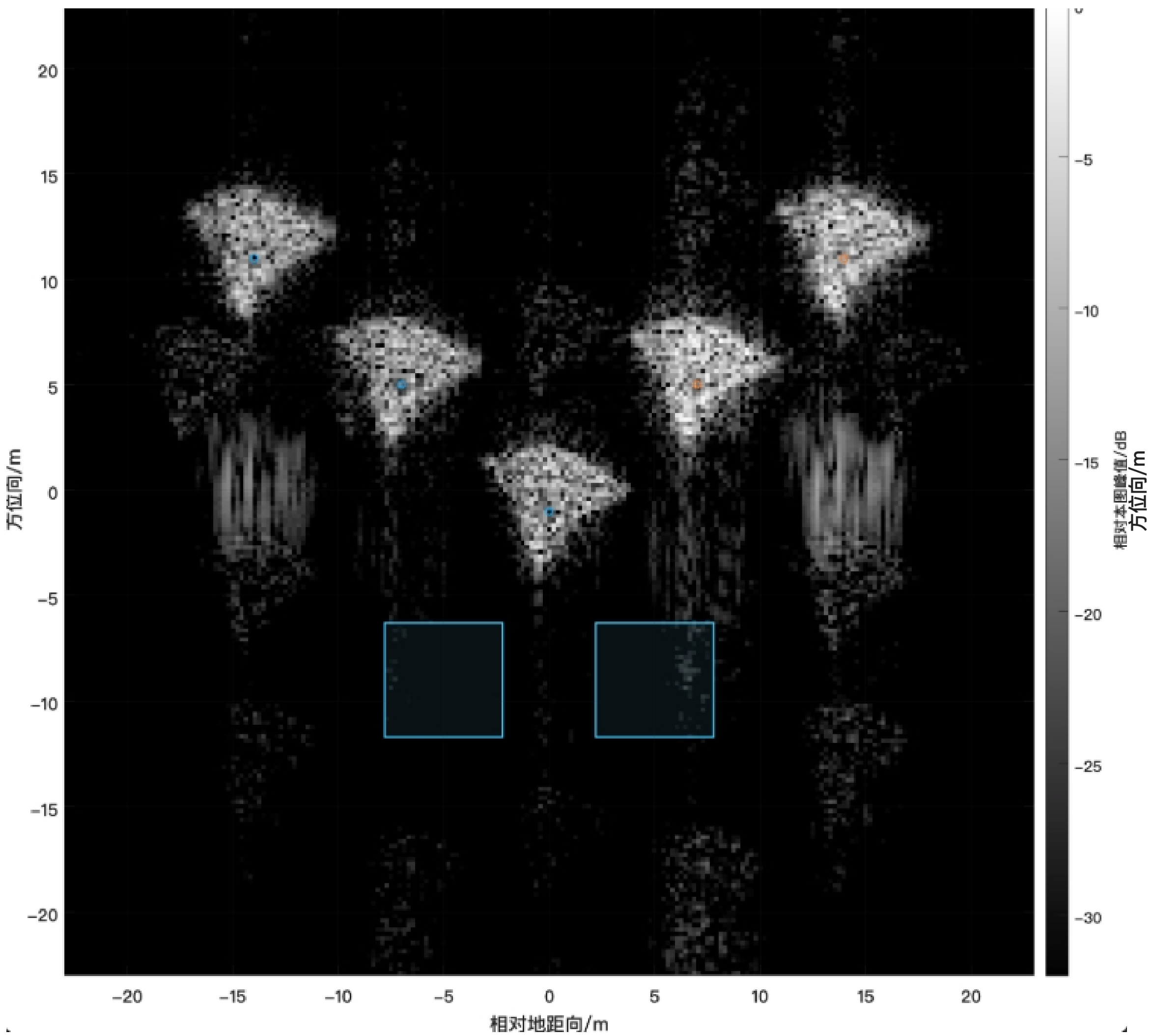


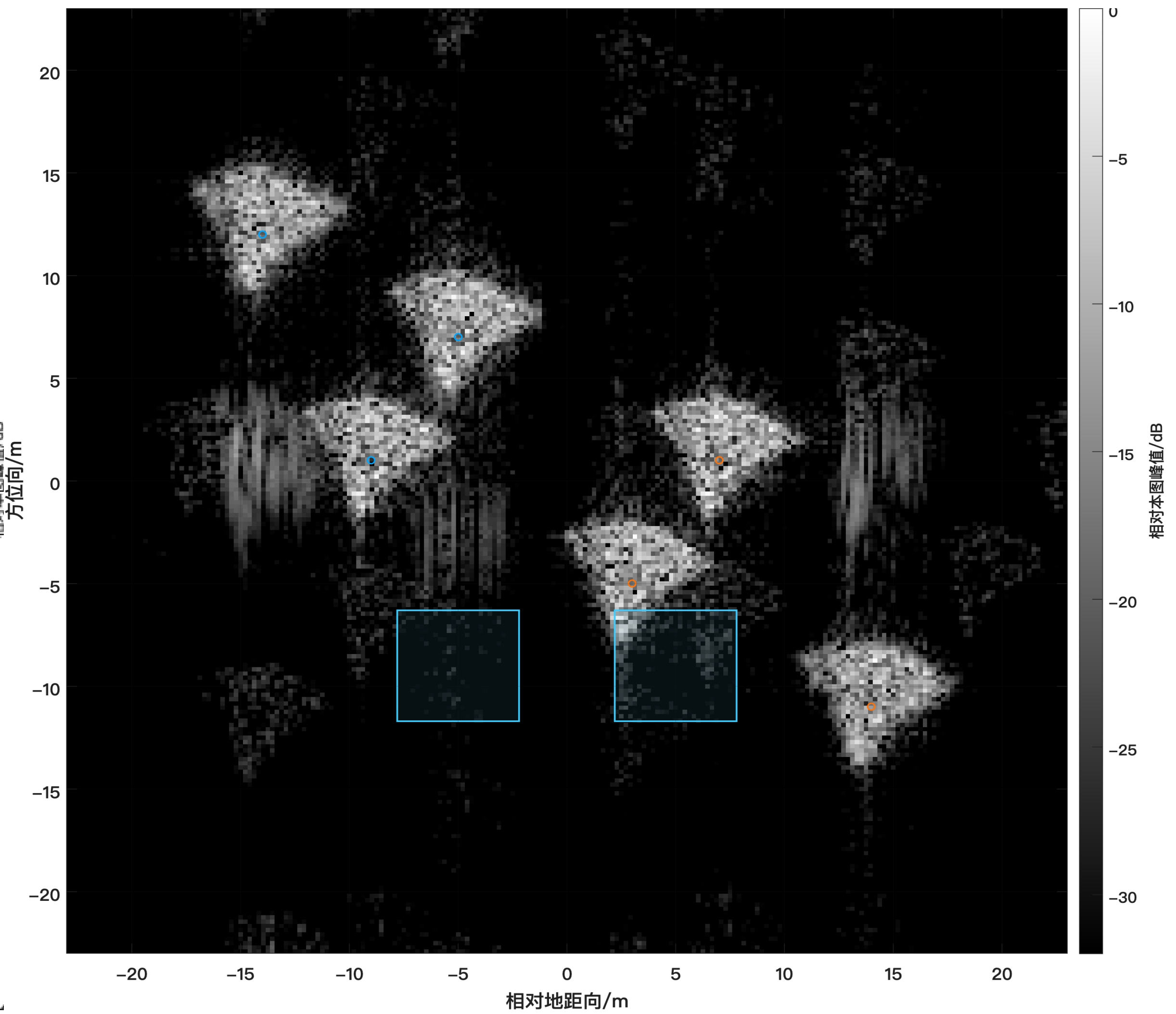

## 2.2 重叠



# 1 Meta Pencil自由度

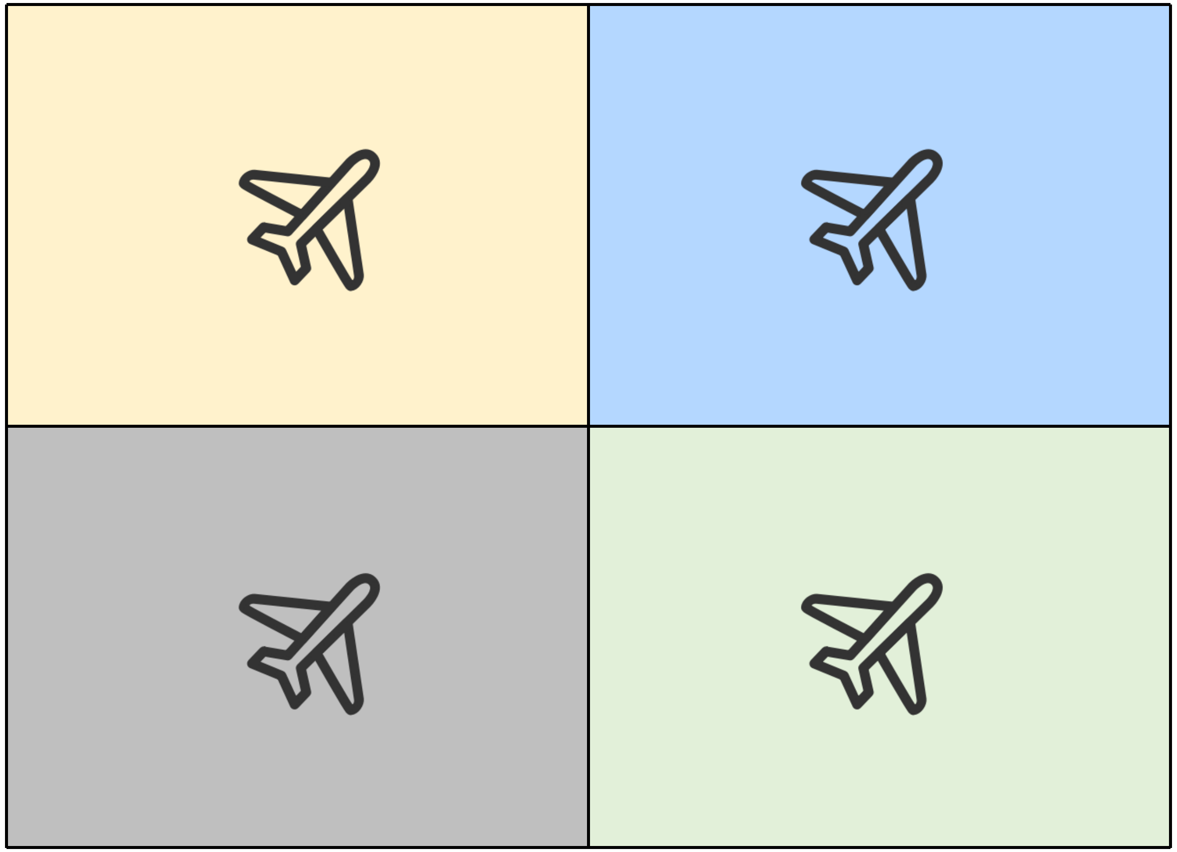
## 2.3 独立



# 1 Meta Pencil自由度

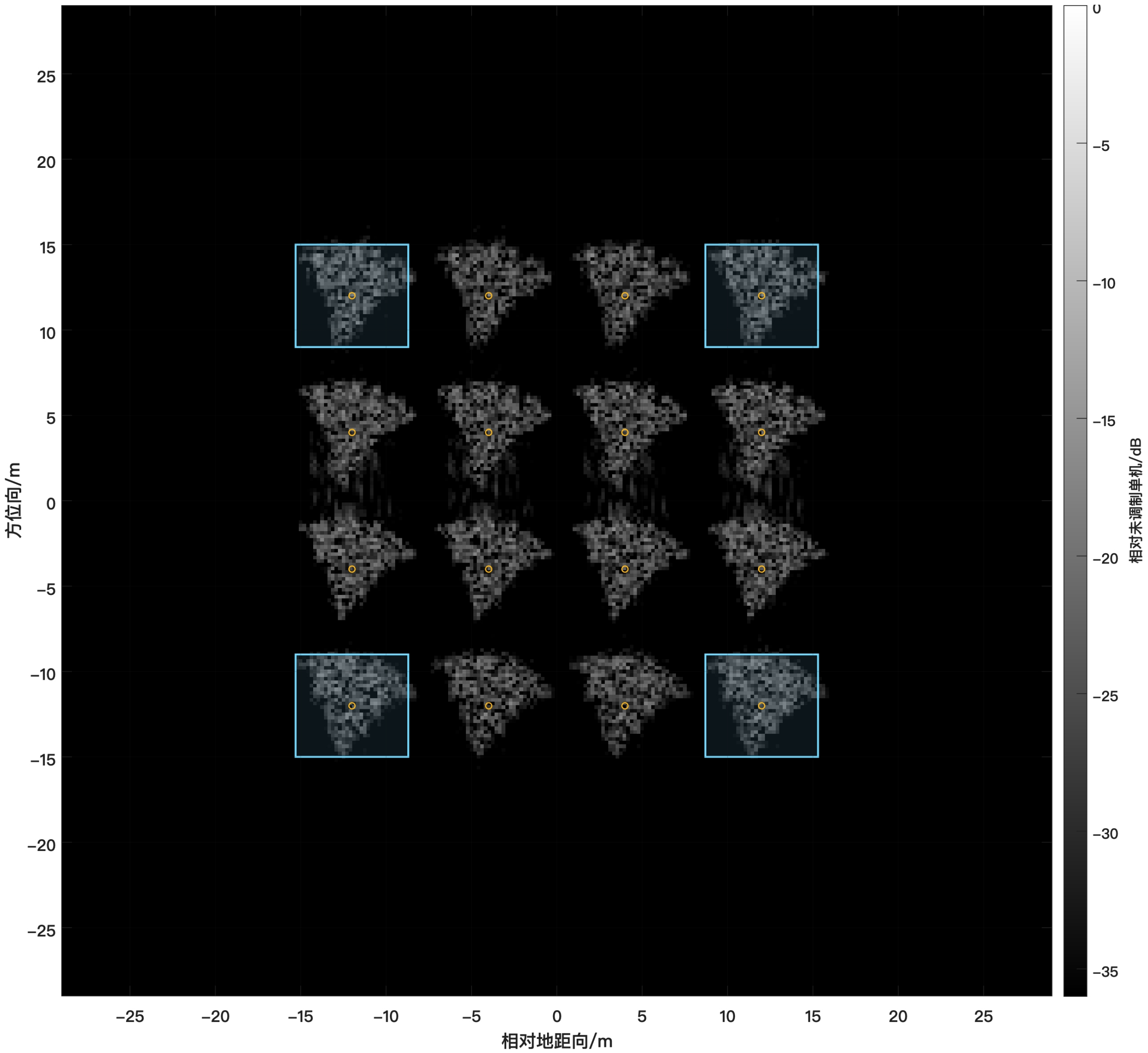
3 四架飞机

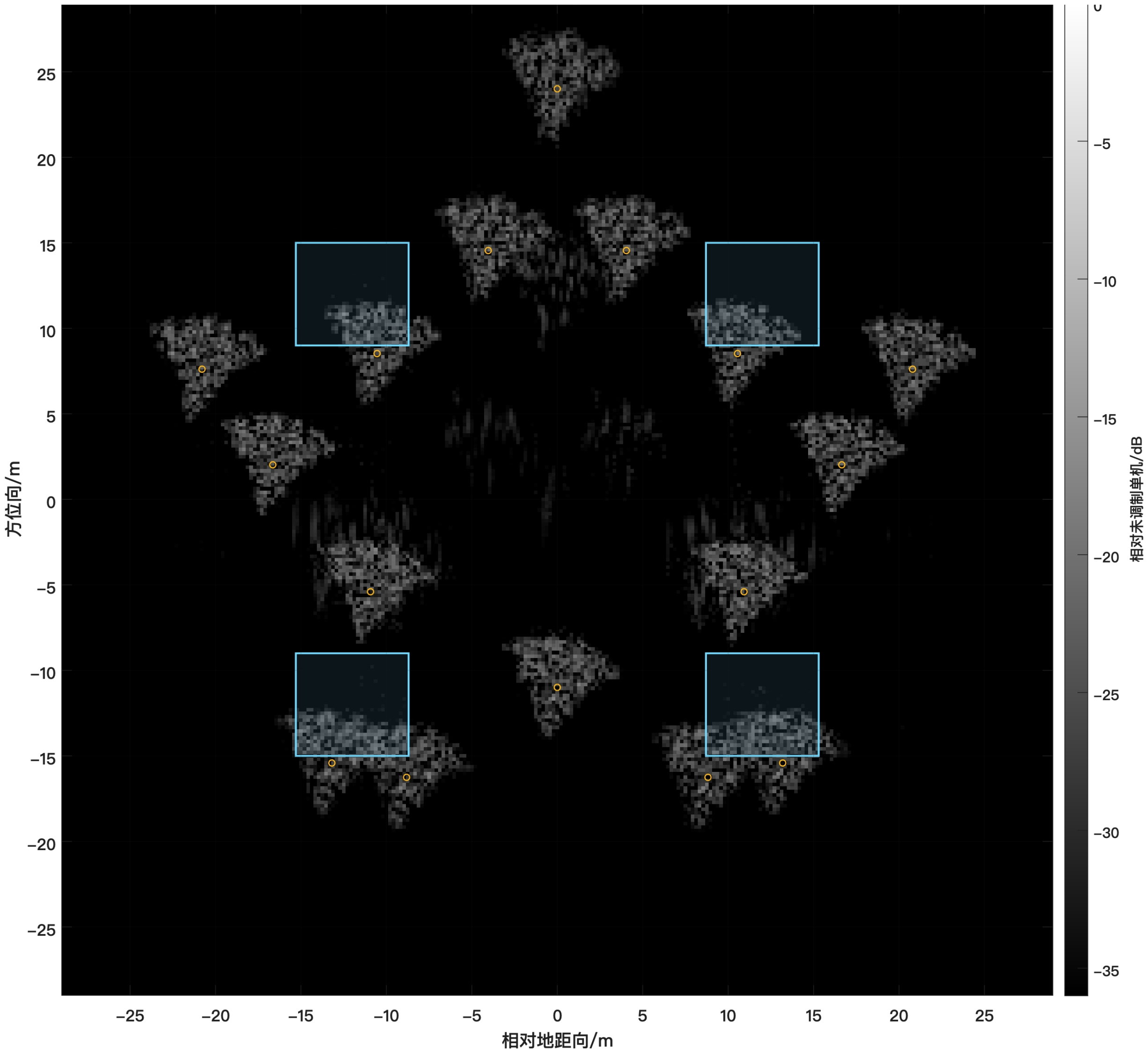


每架飞机独立编码

# 1 Meta Pencil自由度

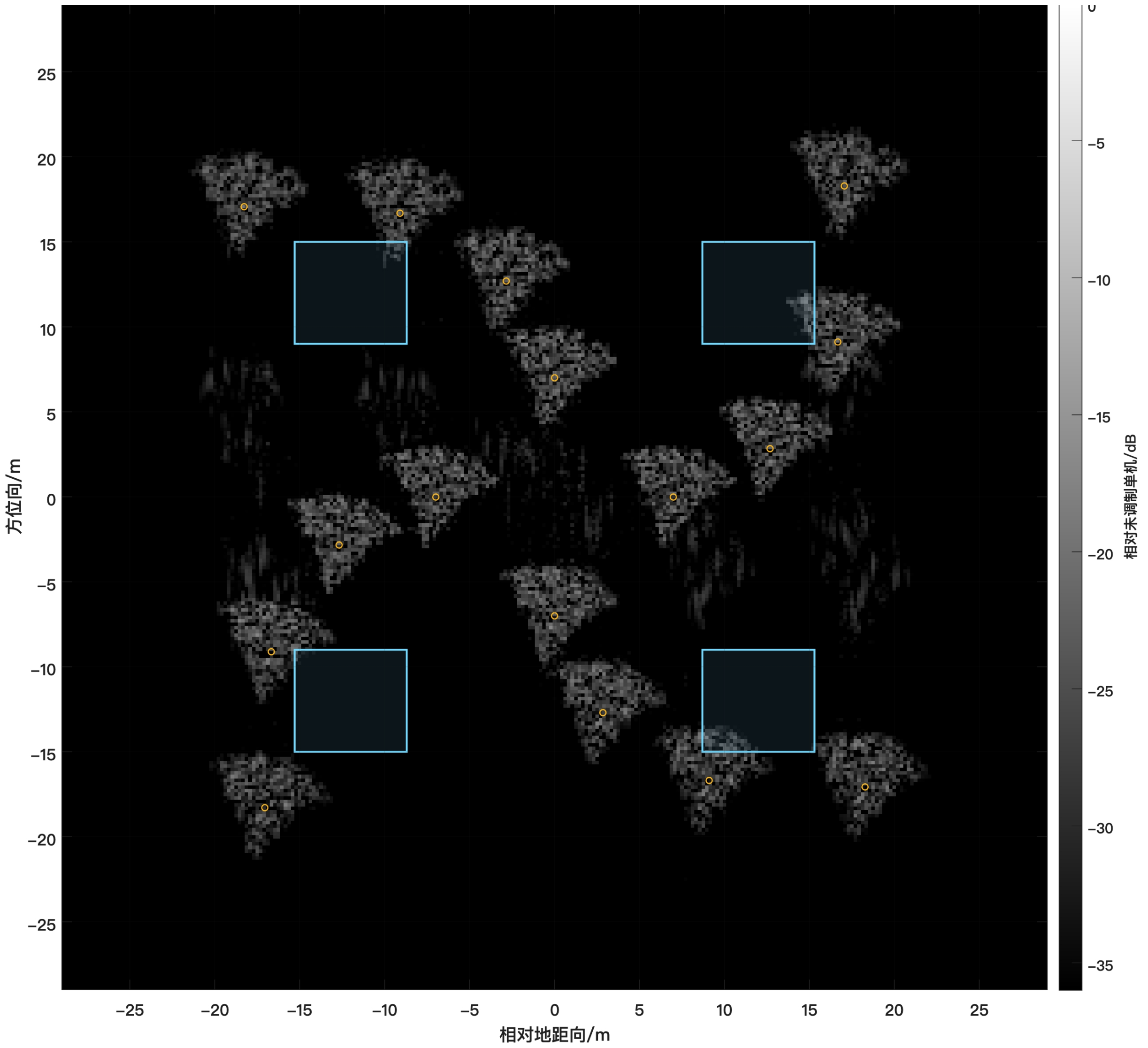
## 3 四架飞机

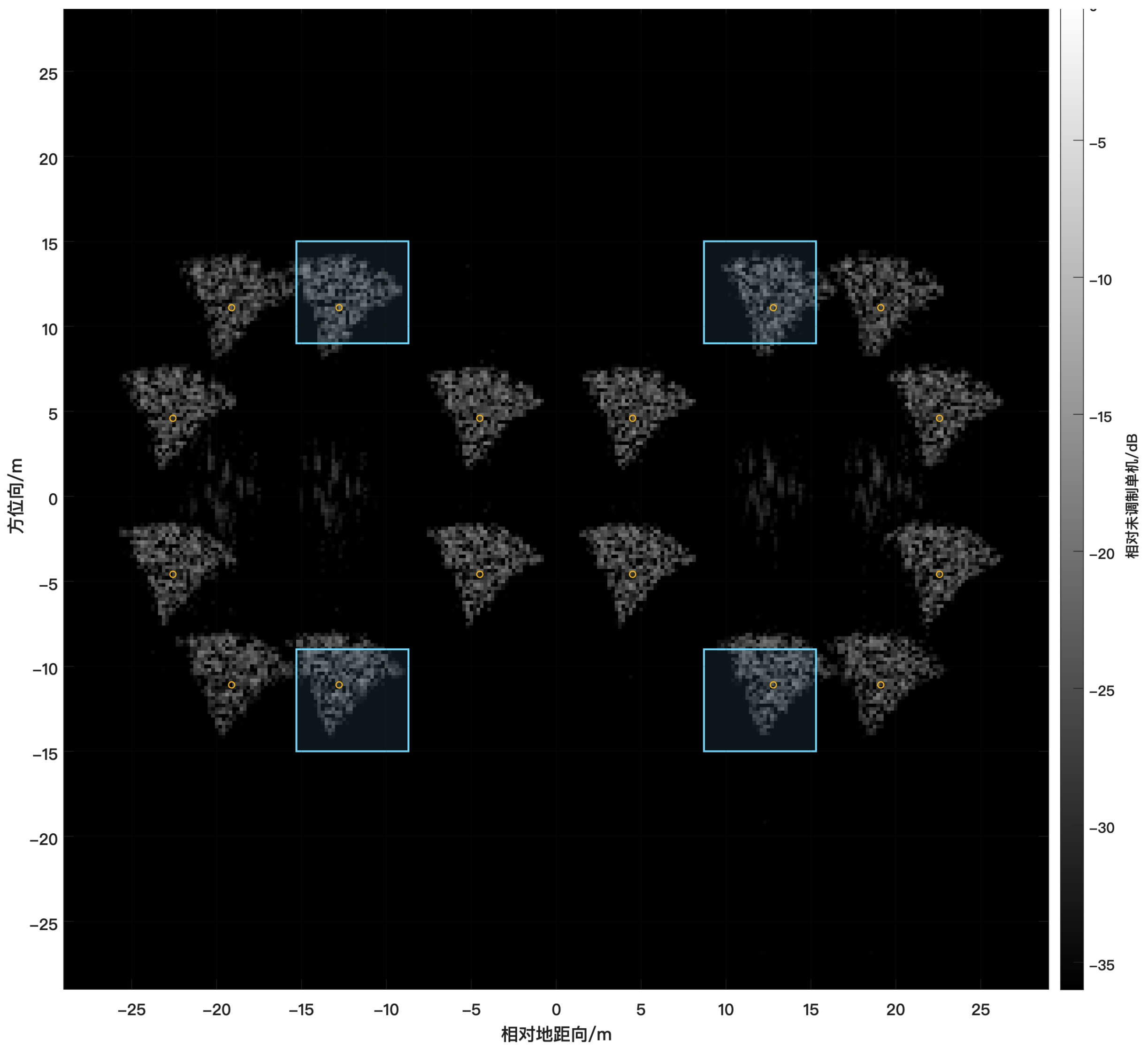


# 1 Meta Pencil自由度

## 3 四架飞机



# 汇报完毕，敬请老师批评指正！

汇报人：谢秋实　　汇报日期：2026 / 08 / 18